

Screening estimates for the actual unweighted disk gas

Boundary attenuation and exactly centered reference gases

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Abstract

We prove screening estimates beyond the determinantal case for the original finite unweighted disk log gas. For every $0 < \beta \leq 2$, a negative-type distance argument gives a particle-number-uniform susceptibility inequality under every common one-body weight. Its consequences include generating-functional bounds and total-variation stability of the complete particle law under perturbations with small quadratic intensity cost. A positive radial-gap mixture supplies a uniform hyperbolic intensity bound at every temperature. Combining these facts, distant Green-potential tilts have vanishing effect on the actual law, uniformly in N , and the full exterior Green force has vanishing mean-square tail. The limiting Green force is therefore recoverable from the limiting exterior configuration. Independently, the nondegenerate outermost-radius boundary statistic decouples from the entire local law at rate $O(N^{-1})$ for every $\beta > 0$. At $\beta = 2$ we prove full selected-exterior attenuation at rate $O(1 - R^2)$, uniformly in N , by reversing the conditioning and computing the distant response to an interior hole. We also construct an exact short-range reference representation, including its count weights, with a uniformly small comparison norm for $0 < \beta \leq 1/4$. For $0 < \beta \leq 1/20$, a convergent complex cluster expansion gives a periodic restoring potential in the exact imaginary Gaussian-field representation. A self-consistent reference then cancels the linear field term exactly while retaining these uniform small-temperature bounds. A separate strongly convex construction gives exact canonical centering at every N and $0 < \beta < 2$. The remaining large-field and normalization problems are not discarded. Appendices give a non-determinantal two-to-one deletion coupling and whole-local-law linear response at $\beta = 2$. Independence from the entire distant exterior away from 2, and the conjectured intensity coefficient, remain unproved.

Contents

1 The actual model and the scope of the results

The finite gas throughout the note is

$$P_{N,\beta}(d\mathbf{z}) = \frac{1}{Z_N(\beta)} \prod_{i < j} |z_i - z_j|^\beta \prod_i \mu(dz_i), \quad \mu = \frac{dA}{\pi}, \quad z_i \in \mathbb{D}, \quad \beta > 0.$$

Write $\Xi_N = \sum_i \delta_{z_i}$ and $B_r = \{|z| < r\}$. Intensities $\rho_{N,\beta}$ are relative to μ . Use curvature -1 hyperbolic distance d and area

$$\nu(dz) = \frac{\mu(dz)}{(1 - |z|^2)^2}, \quad d(0, z) = 2 \operatorname{artanh} |z|.$$

The hyperbolic ball of radius L has ν -volume $\sinh^2(L/2)$. For probabilities, $\operatorname{TV}(P, Q) = \sup_E |P(E) - Q(E)|$ takes values in $[0, 1]$. Thus a TV estimate for a point-process restriction controls all its positions and count probabilities at once.

The full boundary target is, for fixed $r < 1$,

$$\sup_N \mathbb{E} \operatorname{TV} \left(\mathcal{L}(\Xi_N|_{B_r} \mid \Xi_N|_{\mathbb{D} \setminus B_R}), \mathcal{L}(\Xi_N|_{B_r}) \right) \longrightarrow 0 \quad (R \uparrow 1). \quad (1)$$

The results below establish specific parts of this program at fixed temperatures. Theorem ?? proves (??) at $\beta = 2$, including all local counts, for boundary data selected by the actual finite gases. It remains unproved away from the determinantal point. The following distinctions are maintained:

1. The susceptibility and Green-tail estimates use the original canonical gas and hold for the whole interval $0 < \beta \leq 2$.
2. The radial-scale theorem conditions on one nondegenerate statistic measurable from a distant exterior; it does not condition on every detail of that exterior.
3. The small reference norm belongs to a specified short-range reference law. The exact original law retains an additional tilt.
4. Exact self-consistent centering removes the reference's linear field term. The canonical construction does not automatically inherit the grandcanonical influence bound or its phase coercivity.
5. The temperature-response theorem concerns derivatives taken at 2 before $N \rightarrow \infty$; it does not control a fixed nonlinear temperature interval.

The central new input is

$$\operatorname{Var} \Xi_N(f) \leq \mathbb{E} \Xi_N(f^2) - \frac{(\mathbb{E} \Xi_N(f))^2}{N}, \quad 0 < \beta \leq 2. \quad (2)$$

It remains valid under arbitrary common finite one-body weights, including exact finite exterior weights. The proof below is a self-contained Hilbert-space and spectral induction. Its connection with Lorentzian polynomials uses the original theorem of Brändén–Huh [?]; that theorem is not needed for the direct continuum proof. This supplies a new input to the screening program studied in the preceding note, rather than simply repeating a comparison criterion with unverified hypotheses.

All asserted estimates are proved in the following sections. Standard spectral, measure-theoretic, elliptic, and determinantal identities used in the arguments are specified there. The longest independent mathematical checks concerned the spectral induction, the radial transport, the Green-field limit, the exact reference normalization, the all-order signed Janossy inversion, the full-exterior comparison, and the new self-consistent centering constructions.

2 A count-sensitive susceptibility bound for the actual gas below two

This section gives an estimate for the original interaction, uniformly in particle number and in every finite exterior weight. It uses the fact that $|z - w|^\beta$ is a squared Hilbert distance for $0 < \beta \leq 2$. The corresponding finite generating polynomials are Lorentzian. We also give a direct spectral induction proof of the needed consequence, so the continuum estimate does not depend on a discretization argument.

Lemma 2.1 (A Hilbert representation). *For $0 < \beta \leq 2$ there are a real Hilbert space H and a continuous map $\Phi : \mathbb{R}^2 \rightarrow H$ such that*

$$\|\Phi(x) - \Phi(y)\|_H^2 = |x - y|^\beta.$$

Consequently, for any finite points $x_1, \dots, x_q$ and real numbers c_i with $\sum_i c_i = 0$,

$$\sum_{i,j} c_i c_j |x_i - x_j|^\beta \leq 0.$$

Proof. For $\beta = 2$, take $H = \mathbb{R}^2$ and $\Phi(x) = x$. For $0 < \beta < 2$, put

$$C_\beta = \int_{\mathbb{R}^2} |e^{ik \cdot e_1} - 1|^2 |k|^{-2-\beta} dk.$$

This number is finite and strictly positive: near zero the numerator is at most $|k|^2$, and outside the unit ball it is at most four. Use the underlying real Hilbert space of $L^2(\mathbb{R}^2, |k|^{-2-\beta} dk; \mathbb{C})$, and set $\Phi(x)(k) = C_\beta^{-1/2} (e^{ik \cdot x} - 1)$. Rotation invariance and the substitution $u = |x - y|k$ give the displayed distance identity. That identity also proves continuity. Finally, expanding squared Hilbert distances, with $\sum_i c_i = 0$, gives

$$\sum_{i,j} c_i c_j \|\Phi(x_i) - \Phi(x_j)\|^2 = -2 \left\| \sum_i c_i \Phi(x_i) \right\|^2 \leq 0.$$

□

Proposition 2.2 (Lorentzian finite approximations). *Let $x_1, \dots, x_q$ be distinct planar points, let $w_i > 0$, and let $1 \leq n \leq q$. If $0 < \beta \leq 2$, then the polynomial*

$$F_n(t_1, \dots, t_q) = \sum_{\substack{S \subset \{1, \dots, q\} \\ |S|=n}} \left(\prod_{\{i,j\} \subset S} |x_i - x_j|^\beta \right) \prod_{i \in S} w_i t_i$$

is Lorentzian. In particular, $\log F_n$ and $F_n^{1/n}$ are concave on the positive orthant.

Proof. We use the characterization and concavity theorem of Brändén–Huh [?, Theorems 2.25 and 2.30, Proposition 2.33]: a homogeneous polynomial with nonnegative coefficients and M -convex support is Lorentzian if every derivative of degree two has a Hessian with at most one positive eigenvalue; such a polynomial has the two stated concavity properties.

Here the support consists of all zero-one vectors with coordinate sum n . It satisfies the required exchange property: if $i \in S \setminus T$, choose any $j \in T \setminus S$ and exchange i, j . For $n \geq 2$, a nonzero derivative indexed by an $(n - 2)$ -element set A has quadratic coefficients

$$C_A b_i b_j |x_i - x_j|^\beta, \quad b_i = w_i \prod_{a \in A} |x_i - x_a|^\beta, \quad i, j \notin A,$$

with $C_A > 0$. Its Hessian is thus a positive diagonal congruence of the matrix $K_{ij} = |x_i - x_j|^\beta$, with zero diagonal. Lemma ?? makes the quadratic form of K nonpositive on the codimension-one subspace $\sum_i c_i = 0$. The min-max principle implies that K has at most one positive eigenvalue, and diagonal congruence preserves its inertia. Derivatives with repeated indices vanish. This checks every hypothesis. Degree one is immediate. $\square$

Theorem 2.3 (Uniform susceptibility for every finite exterior weight). *Let σ be a nonzero finite nonatomic Borel measure with compact support in $\mathbb{R}^2$, let $n \geq 1$, and let $0 < \beta \leq 2$. Normalize the symmetric probability measure*

$$P_{n,\beta}^\sigma(d\mathbf{z}) = \frac{1}{Z_{n,\beta}^\sigma} \prod_{i < j} |z_i - z_j|^\beta \prod_{i=1}^n \sigma(dz_i).$$

Write $\Xi = \sum_i \delta_{z_i}$ and let ρ be its intensity measure, so $\rho(\mathbb{R}^2) = n$. Every bounded real Borel function f satisfies

$$\text{Var} \Xi(f) \leq \int f^2 d\rho - \frac{1}{n} \left(\int f d\rho \right)^2 \leq \int f^2 d\rho. \quad (3)$$

For every bounded nonnegative Borel function v ,

$$\mathbb{E} \prod_{z \in \Xi} v(z) \leq \left(\frac{1}{n} \int v d\rho \right)^n \leq \exp \left\{ \int (v - 1) d\rho \right\}. \quad (4)$$

Consequently, for every real t ,

$$\log \mathbb{E} e^{t(\Xi(f) - \mathbb{E} \Xi(f))} \leq \int (e^{tf} - 1 - tf) d\rho. \quad (5)$$

These inequalities hold with the same constants after replacing σ by any nonzero finite nonatomic one-body weight.

Direct proof of the variance estimate. Compact support makes the interaction bounded and the partition function finite. Nonatomicity makes the partition function positive, since the particles are pairwise distinct almost everywhere. Normalize σ to have mass one, which does not change the law. We prove the first inequality by induction on n . For $n = 1$ it is an equality.

For $n = 2$, use Lemma ?? and write $m = \int \Phi d\sigma$, $H(z) = \Phi(z) - m$, and $V = \int \|H\|^2 d\sigma > 0$. The pair partition function is $2V$. Let f have mean zero under the tagged-particle marginal, and put

$$a = \int f d\sigma, \quad b = \int f H d\sigma, \quad c = \int f \|H\|^2 d\sigma.$$

The tagged marginal has density $(\|H(z)\|^2 + V)/(2V)$ relative to σ , so the centering condition is $c + Va = 0$. Expanding the Hilbert distance gives

$$\mathbb{E}[f(z_1)f(z_2)] = \frac{2ac - 2\|b\|^2}{2V} = -a^2 - \frac{\|b\|^2}{V} \leq 0.$$

Thus $\text{Var}(f(z_1) + f(z_2)) \leq 2\mathbb{E}f(z_1)^2$, as required for centered f .

Suppose now that the assertion holds for $n - 1 \geq 2$, for every such one-body measure. Let π be the tagged-particle marginal of the n -particle law, and define the self-adjoint Markov operator on $L^2(\pi)$ by

$$(Tf)(x) = \mathbb{E}[f(z_2) \mid z_1 = x].$$

Exchangeability gives self-adjointness, and conditional Jensen gives $\|T\| \leq 1$. Take bounded f with $\pi(f) = 0$, and write

$$M = \pi(f^2), \quad A = \langle f, Tf \rangle_\pi, \quad B = \|Tf\|_{L^2(\pi)}^2.$$

Conditioned on $z_1 = x$, the other $n - 1$ particles have precisely the same interaction and one-body measure $|x - z|^\beta \sigma(dz)$. This measure remains finite and nonatomic. The induction hypothesis and total variance therefore give

$$\begin{aligned} \text{Var} \sum_{i=1}^n f(z_i) &\leq (n-1)(M-B) + \text{Var}(f(z_1) + (n-1)Tf(z_1)) \\ &= nM + 2(n-1)A + (n-1)(n-2)B. \end{aligned}$$

On the other hand, exchangeability gives the exact expression $nM + n(n-1)A$ for the left side. Hence $A \leq B$, since $n > 2$. Bounded functions are dense in $L^2(\pi)$, so on the mean-zero subspace this is the operator inequality $T \leq T^2$. The spectral theorem and $\|T\| \leq 1$ place its spectrum in $[-1, 0] \cup \{1\}$.

The eigenvalue one has only constant eigenfunctions. Indeed,

$$\langle f, (I - T)f \rangle_\pi = \frac{1}{2} \mathbb{E}(f(z_1) - f(z_2))^2,$$

and the pair marginal has a strictly positive density relative to $\sigma \otimes \sigma$ almost everywhere. To see positivity, fix two distinct points; the integral over the other $n - 2$ coordinates is strictly positive because nonatomicity excludes all collisions almost everywhere. Thus equality in the displayed formula forces f to be constant σ -almost everywhere, hence also π -almost everywhere. Since one is isolated from the remaining possible spectrum, its spectral subspace is exactly this eigenspace. There is therefore no positive spectrum on the mean-zero subspace, and $A \leq 0$. It follows that

$$\text{Var} \sum_i f(z_i) = nM + n(n-1)A \leq nM.$$

Subtracting the tagged mean from an arbitrary bounded f proves (??) and completes the induction. $\square$

Generating-functional and Laplace bounds. First suppose that v is bounded and bounded away from zero. Set $v_s = 1 + s(v-1)$, $0 \leq s \leq 1$, and

$$G(s) = \mathbb{E} \prod_{z \in \Xi} v_s(z), \quad L(s) = \log G(s).$$

All derivatives can be taken under the finite integral. Under the tilted gas with one-body measure $v_s \sigma$, put $h_s = (v-1)/v_s$. Exact differentiation gives

$$L'(s) = \mathbb{E}_s \Xi(h_s), \quad L''(s) = \text{Var}_s \Xi(h_s) - \mathbb{E}_s \Xi(h_s^2) \leq -\frac{L'(s)^2}{n},$$

where the last step is (??) for that tilted one-body measure. Consequently $(G^{1/n})'' \leq 0$. Its tangent line at zero gives

$$G(1)^{1/n} \leq 1 + \frac{1}{n} \int (v-1) d\rho = \frac{1}{n} \int v d\rho.$$

Apply this first to $v + \varepsilon$ and let $\varepsilon \downarrow 0$ to include bounded nonnegative v . The exponential inequality follows from $(1 + u/n)^n \leq e^u$ for $u \geq -n$, with the zero case interpreted directly. Finally set $v = e^{tf}$ and subtract $t \int f d\rho$ from the logarithm to obtain (??). $\square$

Corollary 2.4 (Count sectors of the actual conditional gas). *Let U be any bounded observation region, let Y be a finite exterior configuration, and condition the disk gas on having exactly m points in U , with exterior Y . For $0 < \beta \leq 2$, all conclusions of Theorem ?? hold with $n = m$ and*

$$\sigma(dz) = \mathbf{1}_U(z) \prod_{y \in Y} |z - y|^\beta dA(z).$$

The empty sector is deterministic. In any nonempty sector, for a Borel set $A \subset U$ and $a = \mathbb{E}\Xi(A)$,

$$\text{Var } \Xi(A) \leq a(1 - a/m), \quad \mathbb{E}t^{\Xi(A)} \leq \left(1 + \frac{a}{m}(t-1)\right)^m \leq e^{a(t-1)}, \quad t > 0.$$

Proof. The displayed measure is finite and nonatomic, and nonzero whenever the sector is admissible. Apply the theorem with $f = \mathbf{1}_A$ and $v = 1 + (t-1)\mathbf{1}_A$. This proof requires no separation of Y into annuli and no approximation of its entire interaction. $\square$

Remark 2.5 (Scope of the negative dependence). The theorem controls the variance of every real linear statistic and the full generating functional, including count fluctuations. It does not assert that the covariance of two different nonnegative test functions is always nonpositive, nor that arbitrary local events are negatively associated. It does not by itself provide stochastic domination between successive particle numbers. A spatial attenuation estimate still needs a perturbation whose quadratic intensity cost vanishes with distance.

3 Radial comparison and an actual screened boundary statistic

All statements in this section concern the unweighted disk gas at an arbitrary fixed $\beta > 0$. Intensities are relative to $\mu(dz) = dA(z)/\pi$. Set

$$d_j = 2j + \frac{\beta}{2}j(j-1), \quad c_\beta = \min\{2, \beta/2\}, \quad C_\beta = 2 + \frac{16\sqrt{6}}{c_\beta}.$$

The notation c_β here is merely a comparison constant, not the conjectured hyperbolic intensity coefficient.

Proposition 3.1 (A positive mixture for the ordered radial gaps). *Order the radii increasingly as $r_1 < \dots < r_N$, put $t_j = -\log r_j$, $t_{N+1} = 0$, and $g_j = t_j - t_{j+1}$. The joint distribution of $(g_1, \dots, g_N)$ is a countable mixture of product distributions*

$$\bigotimes_{j=1}^N \text{Exp}(d_j + 2A_j), \quad A_j \in \mathbb{Z}_{\geq 0}, \quad A_N = 0.$$

In particular, on a coupling with independent rate-one exponentials E_j , one has simultaneously

$$g_j \leq E_j/d_j, \quad t_j \leq \sum_{k=j}^N E_k/d_k. \quad (6)$$

Proof. Ties and zero radii have probability zero. On the ordered region factor

$$\prod_{i < j} |z_j - z_i|^\beta = \prod_{j=1}^N r_j^{\beta(j-1)} \left| \prod_{i < j} (1 - z_i/z_j)^{\beta/2} \right|^2.$$

Every power is the analytic branch defined by its binomial expansion on $|z_i/z_j| < 1$. The finite product has an absolutely convergent Laurent expansion on every fixed ordered radial torus,

$$\prod_{i < j} (1 - z_i/z_j)^{\beta/2} = \sum_{\alpha} b_{\alpha} \prod_j z_j^{\alpha_j}.$$

An expansion term is specified by nonnegative integers n_{ij} , $i < j$, and has

$$\alpha_j = \sum_{\ell > j} n_{j\ell} - \sum_{i < j} n_{ij}, \quad A_k := \sum_{j \leq k} \alpha_j = \sum_{i \leq k < j} n_{ij} \geq 0, \quad A_N = 0.$$

For a fixed exponent vector, each n_{ij} is bounded by any A_k with $i \leq k < j$, so only finitely many terms contribute to its coefficient. Absolute convergence also justifies regrouping terms. Angular Parseval therefore gives

$$\int \left| \prod_{i < j} (1 - z_i/z_j)^{\beta/2} \right|^2 \prod_j \frac{d\theta_j}{2\pi} = \sum_{\alpha} |b_{\alpha}|^2 \prod_j r_j^{2\alpha_j}.$$

The radial area elements are $2r_j dr_j$. After the changes to t_j and then g_j , the contribution of a term is, up to a common positive constant,

$$|b_{\alpha}|^2 \exp \left\{ - \sum_{k=1}^N (d_k + 2A_k) g_k \right\} \prod_k dg_k.$$

Indeed, the coefficient of g_k is $\sum_{j \leq k} [2 + \beta(j-1) + 2\alpha_j] = d_k + 2A_k$. All rates are positive. Tonelli's theorem applies to this nonnegative series. Its integral is finite and nonzero because it is the original finite partition integral. Normalizing its terms gives mixture weights proportional to $|b_{\alpha}|^2 \prod_k (d_k + 2A_k)^{-1}$ and proves the claim. Finally, sample the mixture index and independent E_j , and set $g_j = E_j/(d_j + 2A_j)$ to obtain (??). $\square$

Theorem 3.2 (Uniform compact intensity bound). *For all $N \geq 1$, $\beta > 0$, and $z \in \mathbb{D}$,*

$$\rho_{N,\beta}(z) \leq \frac{C_{\beta}}{(1 - |z|^2)^2}. \quad (7)$$

No determinantal representation is used.

Proof. The inequality $d_k \geq c_{\beta} k^2$ follows directly from its definition. Independence of the exponentials in (??) gives

$$\mathbb{E} t_j^2 \leq \left(\sum_{k=j}^N d_k^{-1} \right)^2 + \sum_{k=j}^N d_k^{-2} \leq \frac{6}{c_{\beta}^2 j^2}.$$

Here $\sum_{k \geq j} k^{-2} \leq 2/j$ and $\sum_{k \geq j} k^{-4} \leq 2/j^2$ suffice. Since $\{\Xi_N(B_r) \geq j\} = \{t_j > -\log r\}$ up to null events, Markov's inequality yields, for $0 < r < 1$,

$$\mathbb{E} \Xi_N(B_r) \leq \sum_{j \geq 1} \min \left\{ 1, \frac{6}{c_{\beta}^2 j^2 (-\log r)^2} \right\} \leq 1 + \frac{2\sqrt{6}}{c_{\beta} (-\log r)}. \quad (8)$$

For the last inequality, put $a = \sqrt{6}/(c_{\beta} (-\log r))$, split the sum at $\lceil a \rceil$, and use $\sum_{j > m} j^{-2} \leq 1/m$.

The intensity is rotation invariant and subharmonic in its argument: its marginal formula is a positive integral of $|\prod_{i=2}^N (z - z_i)|^{\beta}$ against a measure independent of z . These integrands are subharmonic, and are locally uniformly bounded for fixed N . A radial subharmonic function finite at zero is nondecreasing with radius, as follows from the monotonicity of its circular means. Consequently, whenever $0 \leq r < s < 1$,

$$\rho_{N,\beta}(r)(s^2 - r^2) \leq \int_{B_s \setminus B_r} \rho_{N,\beta} d\mu \leq \mathbb{E} \Xi_N(B_s).$$

Choose $s^2 = (1 + r^2)/2$. Then $-\log s \geq (1 - s^2)/2 = (1 - r^2)/4$. Substitution of (??) gives

$$\rho_{N,\beta}(r) \leq \frac{2}{1 - r^2} + \frac{16\sqrt{6}}{c_{\beta}(1 - r^2)^2} \leq \frac{C_{\beta}}{(1 - r^2)^2}.$$

$\square$

Proposition 3.3 (Exact scale separation). *Let $R_N = \max_i |z_i|$, $W_N = \{z_i/R_N : 1 \leq i \leq N\}$, and $T_N = d_N(-\log R_N)$. Then T_N has the rate-one exponential law and is independent of the whole normalized configuration W_N . The intensity of W_N in the open disk satisfies the same bound (??).*

Proof. For completeness, independence of the whole shape, rather than only its radii, follows from a direct change of variables. Choose the almost surely unique outermost label, write its coordinate as $Re^{i\theta}$, and write all other coordinates as Rw_i , $|w_i| < 1$. The Jacobian and the homogeneous Vandermonde yield a factor $R^{d_N-1} dR$. All remaining factors depend only on $(\theta, w_1, \dots, w_{N-1})$. Thus R has density $d_N R^{d_N-1}$ on $(0, 1)$ independently of the shape, proving the first assertion. Equivalently, the gap mixture has $A_N = 0$.

There is one shape particle on the unit circle. The other particles, conditional on its angle, have density proportional to

$$|\Delta_{N-1}(\mathbf{w})|^\beta \prod_{i=1}^{N-1} |w_i - e^{i\theta}|^\beta \prod_i \mu(dw_i).$$

The interior shape intensity, after averaging the angle, is therefore radial and subharmonic by exactly the marginal argument above. Moreover $R_N \leq 1$ gives, on the defining coupling, $W_N(B_s) \leq \Xi_N(B_s)$ for every $s < 1$. Hence (??) applies to its count expectation. The same monotonicity argument proves its stated intensity bound. For $N = 1$ its interior intensity is zero, so the conclusion still holds. $\square$

Theorem 3.4 (Count-law attenuation of the outermost radial mode). *Fix $0 < r < 1$, let $s^2 = (1 + r^2)/2$, and put $K_{\beta,r} = C_\beta/(1 - s^2)^2$. For $N \geq 2$, with $M_N = \Xi_N(B_r)$, one has*

$$\mathbb{E} \text{TV}(\mathcal{L}(M_N | T_N), \mathcal{L}(M_N)) \leq \frac{4K_{\beta,r}r^2}{d_N - 2} + 2(r/s)^{d_N} = O_{\beta,r}(N^{-2}). \quad (9)$$

This is an actual selected-boundary estimate for every fixed $\beta > 0$ and includes the complete probability distribution of the local count. The boundary statistic T_N remains nondegenerate.

Proof. Let Q_t be the law of $W_N(B_{re^{t/d_N}})$, with the evident interpretation if the radius exceeds one, and let Q_0 be the law at radius r . Exact scale separation says that Q_t is a version of the conditional law of M_N given $T_N = t$. Couple these count variables using the same W_N . If $0 \leq t \leq d_N \log(s/r)$, the coupling inequality and the shape intensity bound give

$$\text{TV}(Q_t, Q_0) \leq \mathbb{P}\{W_N(B_{re^{t/d_N}} \setminus B_r) > 0\} \leq K_{\beta,r}r^2(e^{2t/d_N} - 1).$$

For larger t , use the bound one. Since $d_N > 2$ for $N \geq 2$, integration against the exponential density yields

$$\int_0^\infty \text{TV}(Q_t, Q_0)e^{-t} dt \leq \frac{2K_{\beta,r}r^2}{d_N - 2} + (r/s)^{d_N}.$$

The unconditional law is $\bar{Q} = \int Q_t e^{-t} dt$. Convexity and the triangle inequality imply $\int \text{TV}(Q_t, \bar{Q})e^{-t} dt \leq 2 \int \text{TV}(Q_t, Q_0)e^{-t} dt$, proving the claim. $\square$

The same proof controls the joint vector of counts in any fixed finite collection of concentric disks: replace r^2 in the first term by the sum of their squared radii, use their largest radius in the tail term, and choose s above that largest radius. Counts in finitely many concentric annuli follow by taking differences. This does not show independence from the *entire* distant exterior, whose angular information and remaining count selection are absent from T_N . The result is useful because it separates a rigorously attenuated, nontrivial boundary mode from those still requiring collective screening.

Lemma 3.5 (Uniform exponential compact-count moments). *For $0 < b < 1$ and every integer $j \geq 1$,*

$$\mathbb{P}\{\Xi_N(B_b) \geq j\}, \quad \mathbb{P}\{W_N(B_b) \geq j\} \leq \exp \left\{ 2j - \frac{c_\beta(-\log b)}{2} j^2 \right\}. \quad (10)$$

In particular, for every fixed $v > 0$,

$$\sup_N \mathbb{E}[\Xi_N(B_b) e^{v\Xi_N(B_b)}] < \infty, \quad \sup_N \mathbb{E}[W_N(B_b) e^{vW_N(B_b)}] < \infty.$$

Proof. For $j \leq N$ put $q = c_\beta j^2/2$. Then $q/d_k \leq j^2/(2k^2) \leq 1/2$ for $k \geq j$. The gap comparison and $-\log(1-x) \leq 2x$ on $[0, 1/2]$ yield

$$\log \mathbb{E} \exp \left(q \sum_{k=j}^N E_k/d_k \right) \leq 2q \sum_{k=j}^N d_k^{-1} \leq 2j.$$

Exponential Markov proves the first tail bound; it is automatic for $j > N$. The shape bound follows from its coupled count domination. For either count K , monotone summation gives

$$\mathbb{E}[K e^{vK}] = \sum_{j \geq 1} (j e^{vj} - (j-1) e^{v(j-1)}) \mathbb{P}(K \geq j) \leq \sum_{j \geq 1} j \exp \left\{ (v+2)j - \frac{c_\beta(-\log b)}{2} j^2 \right\} < \infty.$$

□

Theorem 3.6 (Full local-law attenuation of the radial scale mode). *For every fixed $\beta > 0$ and $0 < r < 1$, there are constants $A_{\beta,r} < \infty$ and $a_{\beta,r} > 0$, independent of N , such that*

$$\mathbb{E} \text{TV}(\mathcal{L}(\Xi_N|_{B_r} | T_N), \mathcal{L}(\Xi_N|_{B_r})) \leq A_{\beta,r} \frac{N}{d_N} + 2 \exp\{-a_{\beta,r} d_N/N\} = O_{\beta,r}(N^{-1}). \quad (11)$$

This concerns the complete local law, including all particle positions and particle-number probabilities.

Proof. Choose $r < u < b < 1$ and a smooth radial function χ equal to one on B_u and zero outside B_b . Put $V(z) = \chi(z)z$ and $L = \max\{1, \text{Lip}(V)\}$. For

$$0 \leq \eta \leq \eta_0 := \min \left\{ \frac{1}{2L}, \frac{u-r}{2u}, \frac{1}{2} \right\}, \quad F_\eta(z) = z - \eta V(z),$$

the map F_η is a smooth diffeomorphism of the plane, equal to the identity outside B_b . Indeed, its derivative has singular values between $1 - \eta L$ and $1 + \eta L$, the map is globally injective by the same Lipschitz bounds, and it is proper; local invertibility and properness give surjectivity. It consequently preserves B_b and $\mathbb{D}$. Its pair distortions and Jacobian satisfy

$$\left| \log \frac{|F_\eta(x) - F_\eta(y)|}{|x - y|} \right| \leq 2L\eta \quad (x \neq y),$$

$$|\log \det DF_\eta(x)| \leq 4L\eta.$$

The determinant is positive, by continuity from $\eta = 0$. Pair distortions vanish when both points lie outside B_b , and the Jacobian is one there.

Use the shape coordinates from Proposition ??: one point is $e^{i\theta}$ on the unit circle and the remaining $N-1$ points have a density $p_\theta(\mathbf{w})$. The transport fixes the boundary point. Write $K = W_N(B_b)$ and $B = L(2\beta + 4)$. There are at most NK affected pairs and at most K nontrivial one-particle Jacobians. Therefore, almost surely,

$$H_\eta := \log \frac{p_\theta(F_\eta \mathbf{w}) J_{F_\eta}(\mathbf{w})}{p_\theta(\mathbf{w})} \quad \text{satisfies} \quad |H_\eta| \leq B\eta NK.$$

The densities vanish only on collision sets of measure zero, which do not affect this formula. Averaging the usual change-of-variables identity for total variation over θ gives

$$\text{TV}(\mathcal{L}(F_\eta W_N), \mathcal{L}(W_N)) \leq \frac{1}{2} \mathbb{E}|e^{H_\eta} - 1| \leq \frac{1}{2} B\eta N \mathbb{E}[K e^{B\eta N K}].$$

This inequality first holds for the labelled shape coordinates and then for the unordered process by data processing. By Lemma ??, the constant $D = \sup_N \mathbb{E}[K e^K]$ is finite. Whenever $B\eta N \leq 1$, the last display is at most $BD\eta N/2$.

Because F_η is radial, injective, and equals multiplication by $1 - \eta$ on B_u , while $(1 - \eta)u > r$, its restriction satisfies the exact configuration identity

$$(F_\eta W_N)|_{B_r} = ((1 - \eta)W_N)|_{B_r}.$$

Let Q_t now denote the law of $(e^{-t/d_N} W_N)|_{B_r}$ and let Q_0 be the local shape law. Set $a = \min\{\eta_0, 1/B\}$. For $0 \leq t \leq ad_N/N$, $\eta = 1 - e^{-t/d_N} \leq t/d_N$ satisfies both required smallness conditions, and hence

$$\text{TV}(Q_t, Q_0) \leq \frac{BDN}{2d_N} t.$$

For larger t use one. Thus

$$\int_0^\infty \text{TV}(Q_t, Q_0) e^{-t} dt \leq \frac{BDN}{2d_N} + e^{-ad_N/N}.$$

Exact scale separation identifies Q_t as the conditional local law given $T_N = t$. Comparing with its exponential mixture and using the same triangle/convexity argument as in Theorem ?? proves (??), with $A_{\beta,r} = BD$ and $a_{\beta,r} = a$. $\square$

This really is a boundary statistic. For any fixed $r < R < 1$, define $T_{N,R} = T_N$ if $R_N > R$ and set $T_{N,R} = +\infty$ otherwise. It is measurable from the exterior configuration in $\mathbb{D} \setminus B_R$ and is a deterministic function of T_N . Data processing therefore preserves both attenuation bounds when T_N is replaced by $T_{N,R}$. The censoring probability is exactly R^{d_N} , so this selected exterior statistic has an asymptotically nondegenerate exponential distribution. The theorem does not assert that it is a sufficient statistic for the entire exterior.

4 Fixed-temperature screening estimates for the actual gas

We now combine the susceptibility estimate with the radial bound. Every statement in this section uses the original canonical disk law; no effective pair potential replaces its interaction. Write $\nu(dz) = dA(z)/[\pi(1 - |z|^2)^2]$ and retain the intensity constant C_β from Theorem ??.

Theorem 4.1 (Full-law stability under a one-body perturbation). *Let $0 < \beta \leq 2$, let f be bounded and real, and put*

$$\frac{dP_{N,\beta}^{t,f}}{dP_{N,\beta}} = \frac{e^{t\Xi_N(f)}}{\mathbb{E}_{N,\beta} e^{t\Xi_N(f)}}, \quad M = \|f\|_\infty, \quad Q_0 = \mathbb{E}_{N,\beta} \Xi_N(f^2).$$

For every real t ,

$$D_{\text{KL}}(P_{N,\beta}^{t,f} \| P_{N,\beta}) \leq \frac{t^2}{2} e^{|t|M} Q_0, \tag{12}$$

$$\text{TV}(P_{N,\beta}^{t,f}, P_{N,\beta}) \leq \frac{|t|}{2} e^{|t|M/2} \sqrt{Q_0} \leq \frac{|t|\sqrt{C_\beta}}{2} e^{|t|M/2} \|f\|_{L^2(\nu)}. \tag{13}$$

The last inequality is useful when $f \in L^2(\nu)$. The first two apply more generally to every nonzero finite nonatomic one-body measure supported in a bounded region, with its own value of Q_0 . The TV estimate concerns the entire N -particle configuration, and therefore every local count and position event simultaneously.

Proof. Let $P_s = P_{N,\beta}^{s,f}$ and $Q(s) = \mathbb{E}_s \Xi_N(f^2)$. Differentiation is justified because $\Xi_N(f)$ is bounded for fixed N . Theorem ?? applies to every P_s . Thus

$$|Q'(s)| = |\text{Cov}_s(\Xi_N(f^2), \Xi_N(f))| \leq \sqrt{\mathbb{E}_s \Xi_N(f^4) \mathbb{E}_s \Xi_N(f^2)} \leq MQ(s).$$

Gronwall's inequality gives $Q(s) \leq e^{|s|M} Q(0)$. For $\psi(s) = \log \mathbb{E}_0 e^{s \Xi_N(f)}$, this implies $\psi''(s) = \text{Var}_s \Xi_N(f) \leq e^{|s|M} Q_0$. The relative entropy is $t\psi'(t) - \psi(t)$, since $\psi(0) = 0$. Its derivative is $t\psi''(t)$; integration from zero to t , also when $t < 0$, proves (?). Pinsker's inequality with $\text{TV} = \sup_E |P(E) - Q(E)|$ gives $\text{TV}^2 \leq D_{\text{KL}}/2$. Finally, $Q_0 \leq C_\beta \int f^2 d\nu$ by the actual intensity bound. $\square$

Corollary 4.2 (The mean measure is controlled as well). *For a relatively compact Borel window A , put $m_0 = \mathbb{E}_{N,\beta} \Xi_N(A)$ and $D = \frac{|t|}{2} e^{|t|M/2} \sqrt{Q_0}$. The variation of the two mean measures restricted to A is at most*

$$2\sqrt{m_0} D + D^2.$$

Here variation is the supremum of the integral difference over all real u supported in A with $|u| \leq 1$.

Proof. Write $m(s) = \mathbb{E}_s \Xi_N(A)$. The susceptibility estimate gives $|m'(s)| \leq \sqrt{m(s)Q(s)}$. Applying the inequality first to $\sqrt{m(s)} + \epsilon$ and then letting $\epsilon \downarrow 0$ shows $\sqrt{m(s)} \leq \sqrt{m_0} + \frac{1}{2} \int_0^{|s|} \sqrt{Q(\pm v)} dv$, where the sign follows that of s . For u as stated, $|\frac{d}{ds} \mathbb{E}_s \Xi_N(u)| \leq \sqrt{m(s)Q(s)}$. Integrate this bound and use $\frac{1}{2} \int_0^{|t|} \sqrt{Q(\pm v)} dv \leq D$. The result is $2\sqrt{m_0} D + D^2$, independently of u . $\square$

4.1 A distant Green tail may have large mean but small effect

Let

$$g(a, z) = -\log \left| \frac{a - z}{1 - a\bar{z}} \right|.$$

Fix $0 < r < 1$, let σ be a finite real signed source measure supported in $\bar{B}_r$, and write $q = |\sigma|(\bar{B}_r)$. For $R > r$ put

$$V_R(z) = \mathbf{1}_{\{|z| > R\}} \int g(a, z) \sigma(da), \quad \delta = 1 - R^2.$$

This is a potential applied only to the distant particles; t below is its fixed amplitude, not an N -dependent temperature change.

Theorem 4.3 (Full-law attenuation of distant Green-potential tilts). *Suppose $0 < \beta \leq 2$ and $\delta \leq (1 - r)^2/2$. For every $N \geq 1$, every real t , and every source just described,*

$$\text{TV}(P_{N,\beta}^{t,V_R}, P_{N,\beta}) \leq \frac{|t|q\sqrt{C_\beta}}{2(1-r)^2} \exp\left\{ \frac{|t|q\delta}{2(1-r)^2} \right\} \sqrt{\delta}. \quad (14)$$

The same rate controls every local law, including every particle count. On each fixed compact window it also controls the mean measure at order $O_{\beta,r,t,q,A}(\sqrt{\delta})$.

Proof. The identity

$$g(a, z) = -\frac{1}{2} \log(1 - u), \quad u = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - a\bar{z}|^2}$$

and the assumptions give $0 \leq u \leq 1/2$. Since $-\frac{1}{2} \log(1 - u) \leq u$ on this interval,

$$|V_R(z)| \leq \frac{q}{(1-r)^2} (1 - |z|^2) \mathbf{1}_{\{|z| > R\}}.$$

Consequently

$$\|V_R\|_\infty \leq \frac{q\delta}{(1-r)^2}, \quad \int V_R^2 d\nu \leq \frac{q^2\delta}{(1-r)^4},$$

because $(1 - |z|^2)^2 d\nu = d\mu$ and $\mu(\{|z| > R\}) = \delta$. Apply Theorem ??; restriction is a measurable map and cannot increase TV. The mean-measure assertion follows from Corollary ?? and $m_0 \leq C_\beta \nu(A)$. $\square$

The estimate does not follow from a small supremum norm alone: the elementary energy bound would be $N\|V_R\|_\infty$, which is not uniform in N . Moreover $\int |V_R| d\nu$ need not be finite. The proof removes the deterministic mean through normalization and controls the fluctuating sum by its quadratic intensity cost. In curvature -1 , $\sqrt{\delta} = \text{sech}(d(0, R)/2)$; the rate is exponential in the hyperbolic distance at fixed source region. No boundary-volume factor has been omitted.

4.2 The entire distant Green force vanishes in mean square

For $|a| < R$ define the complex force

$$\mathcal{E}_{N,R}(a) := \sum_{|z_i| > R} \frac{1 - |z_i|^2}{(z_i - a)(1 - a\bar{z}_i)} = 2\partial_a \sum_{|z_i| > R} g(a, z_i).$$

This is the field from all particles beyond R , without a truncation of its angular modes.

Theorem 4.4 (Screening and reconstruction of the Green force). *For $0 < \beta \leq 2$, $0 < r < 1$ and $R \geq (1+r)/2$,*

$$\mathbb{E}\mathcal{E}_{N,R}(a) = 0, \quad \sup_{|a| \leq r} \mathbb{E}|\mathcal{E}_{N,R}(a)|^2 \leq \frac{4C_\beta(1-R^2)}{(1-r)^4}, \quad N \geq 1. \quad (15)$$

For every fixed $r < 1$ there is also $C'_{\beta,r} < \infty$ such that

$$\mathbb{E} \sup_{|a| \leq r} |\mathcal{E}_{N,R}(a)|^2 \leq C'_{\beta,r}(1-R^2)$$

for all sufficiently large R , uniformly in N .

Let Ξ_{N_j} converge locally in law to a subsequential limit Ξ . For any $0 < s < 1$, the exterior force

$$\mathcal{E}_\Xi^s(a) = \lim_{R \uparrow 1} \sum_{z \in \Xi: s < |z| < R} \frac{1 - |z|^2}{(z - a)(1 - a\bar{z})}, \quad |a| < s, \quad (16)$$

exists in mean square, locally uniformly in a . It is measurable from the exterior configuration $\Xi|_{\mathbb{D} \setminus B_s}$. Moreover the finite-gas exterior fields $\mathcal{E}_{N_j,s}$ converge jointly with the configurations to this field. Thus this Green force carries no additional random contribution from infinity.

Proof. For a fixed radius $u > |a|$, angular averaging gives

$$\frac{1}{2\pi} \int_0^{2\pi} g(a, ue^{i\theta}) d\theta = -\log u.$$

Indeed the mean of $\log|a - ue^{i\theta}|$ is $\log u$, and that of $\log|1 - aue^{-i\theta}|$ is zero, by their convergent harmonic series. Differentiation in a and radially of the finite-gas intensity prove the zero mean in (??).

For $|a| \leq r$ and $|z| > R \geq (1+r)/2$, the force summand $f_a(z)$ satisfies

$$|f_a(z)| \leq \frac{2(1 - |z|^2)}{(1-r)^2}.$$

Apply (??) to its real and imaginary parts and add the bounds. The zero mean then gives

$$\mathbb{E}|\mathcal{E}_{N,R}(a)|^2 \leq \int_{|z|>R} |f_a(z)|^2 \rho_{N,\beta}(z) d\mu(z),$$

which proves (??) by the intensity bound. Choose $r < r_1 < 1$. The field is holomorphic on B_R . The disk mean-value inequality for its squared modulus implies

$$\sup_{B_r} |\mathcal{E}_{N,R}|^2 \leq \frac{4}{(r_1 - r)^2} \int_{B_{r_1}} |\mathcal{E}_{N,R}(a)|^2 d\mu(a).$$

For $R \geq (1 + r_1)/2$, the already proved estimate and Tonelli give the locally uniform bound. The identical proof applies to a finite radial annulus $R < |z| < R'$ and gives the same upper bound with R .

We detail passage to the point-process limit. On a compact window, (??) with $v = e^{u1_A}$ and (??) bound every exponential count moment uniformly in N . Local convergence therefore passes expectations of continuous compactly supported linear statistics, and their moments, to the limit. In particular, the limiting mean measure is rotation invariant and bounded by $C_\beta \nu$; each fixed circle has zero mean mass and hence almost surely contains no points. These observations justify restriction to each annulus $s < |z| < R$ and convergence of its force, locally uniformly in a . One may first work with smooth radial cutoffs and then remove them using the uniform count-moment and intensity bounds.

The annular mean-square estimate just proved passes to the limit. For every $r < s$, its locally uniform version makes the sums in (??) Cauchy in $L^2(\sup_{B_r} |\cdot|)$. Their limit is a measurable function of the exterior points. Choose a countable sequence $r \uparrow s$; the limits agree on overlaps and define the asserted holomorphic field. Finally, truncate both the finite-gas and limiting fields at radius R . Their joint convergence with configurations holds for fixed R , while the discarded tails have uniformly vanishing mean-square norms as $R \uparrow 1$. The triangle inequality for bounded Lipschitz tests proves the stated joint convergence. $\square$

Integrating this force from zero to a shows that the Green potential modulo its value at zero is recovered as

$$H_\Xi^s(a) = \lim_{R \uparrow 1} \sum_{z \in \Xi: s < |z| < R} [g(a, z) - g(0, z)], \quad |a| < s,$$

with the same local mean-square convergence. Every finite annular sum has mean zero by radiality. This removes the need to specify a separate random mark for this particular Green force. It does not identify the additive potential constant, the conditional count weights, or the harmonic part of the original Euclidean interaction. Indeed

$$-\log |a - z| = g(a, z) - \log |1 - a\bar{z}|,$$

and the second term has boundary modes that the present estimate does not suppress. Consequently Theorem ?? is not a proof of independence from the complete distant exterior sigma-field.

5 The full actual exterior at the determinantal benchmark

The Green-tail tilt estimates do not themselves bound conditioning on the entire far exterior. At $\beta = 2$, however, that stronger statement has a direct proof. It is uniform in the finite particle number, and therefore handles exterior data chosen by the actual gases even along triangular arrays of radii and particle numbers.

For random elements X, Y of standard Borel spaces, write

$$\mathcal{B}(X, Y) = d_{\text{TV}}(\text{Law}(X, Y), \text{Law}(X) \otimes \text{Law}(Y)),$$

where probability total variation takes values in $[0, 1]$. The disintegration formula for total variation gives

$$\mathcal{B}(X, Y) = \mathbb{E} d_{\text{TV}}(\text{Law}(X | Y), \text{Law}(X)) = \mathbb{E} d_{\text{TV}}(\text{Law}(Y | X), \text{Law}(Y)). \quad (17)$$

For clarity, this identity follows by disintegrating the signed measure $\text{Law}(X, Y) - \text{Law}(X) \otimes \text{Law}(Y)$ over either marginal. A countable generating algebra for the standard Borel space, followed by monotone approximation, makes the fiberwise variation measurable and identifies the variation of the signed measure with its integral. Consequently the order of conditioning in (??) can be reversed without changing the dependence coefficient.

Theorem 5.1 (Entire selected-exterior attenuation at two). *Let Ξ_N have density proportional to $|\Delta_N|^2$ relative to μ^N , where $\mu = dA/\pi$ on the unit disk. Fix $0 < r < R < 1$, and put $X_N = \Xi_N|_{B_r}$ and $Y_{N,R} = \Xi_N|_{\mathbb{D} \setminus B_R}$. Then*

$$\mathcal{B}(X_N, Y_{N,R}) \leq \min \left\{ 1, 2 \sum_{j=1}^N \frac{r^{2j}(1 - R^{2j})}{1 - r^{2j}} \right\} \quad (18)$$

$$\leq 2(1 - R^2) \sum_{j=1}^{\infty} \frac{jr^{2j}}{1 - r^{2j}} \leq \frac{2r^2(1 - R^2)}{(1 - r^2)^3}. \quad (19)$$

In particular,

$$\limsup_{R \uparrow 1} \mathbb{E} d_{\text{TV}}(\text{Law}(\Xi_N|_{B_r} | \Xi_N|_{\mathbb{D} \setminus B_R}), \text{Law}(\Xi_N|_{B_r})) = 0.$$

This controls the whole local configuration, including its particle number, and every measurable statistic of the whole far exterior.

We record a proof of the finite-projection comparison needed here. It is a special case of the ordered-kernel theorem [?, Theorem 3.8]; its proof below removes any need for an external coupling theorem in the quantitative argument.

Lemma 5.2 (Finite projection comparison). *Let $H_1 \subset H_2$ be finite-dimensional subspaces of $L^2(E, \mu)$, and let Q_i be their projection determinantal laws. Then $\mathbb{E}_{Q_1} F \leq \mathbb{E}_{Q_2} F$ for every bounded measurable function F increasing under inclusion of configurations. For every measurable $B \subset E$,*

$$d_{\text{TV}}(Q_1|_B, Q_2|_B) \leq \mathbb{E}_{Q_2} \Xi(B) - \mathbb{E}_{Q_1} \Xi(B). \quad (20)$$

Proof. It suffices first to compare ranks n and $n + 1$, with orthonormal basis $\phi_1, \dots, \phi_n$ of H_1 and unit vector $\phi \perp H_1$ such that $H_2 = H_1 \oplus \mathbb{C}\phi$. In the antisymmetric subspace of $L^2(E^n, \mu^n)$, the normalized wave function

$$\Psi(z_1, \dots, z_n) = \frac{1}{\sqrt{n!}} \det[\phi_i(z_j)]_{i,j=1}^n$$

has squared modulus equal to the ordered density of Q_1 . For an antisymmetric function f of n variables define

$$(c_\phi f)(z_1, \dots, z_{n+1}) = \frac{1}{\sqrt{n+1}} \sum_{i=1}^{n+1} (-1)^{i-1} \phi(z_i) f(z_1, \dots, \hat{z}_i, \dots, z_{n+1}),$$

where the hat means that the coordinate is omitted. Its adjoint on antisymmetric functions is the operator

$$(a_\phi f)(z_1, \dots, z_{n-1}) = \sqrt{n} \int_E \overline{\phi(t)} f(t, z_1, \dots, z_{n-1}) d\mu(t).$$

Expanding the finite sum and using antisymmetry gives

$$a_\phi c_\phi + c_\phi a_\phi = I, \quad \|c_\phi f\|_2^2 = \|f\|_2^2 - \|a_\phi f\|_2^2 \leq \|f\|_2^2.$$

For example, in the first identity the term inserting t in the first coordinate contributes f , while the remaining n terms are exactly $-c_\phi a_\phi f$ after their permutation signs are collected. The norm identity follows by taking the inner product with f and using adjointness. These calculations also apply to $n = 0$ with the scalar vacuum space and zero annihilation operator.

Orthogonality implies $a_\phi \Psi = 0$. The normalized wave function of Q_2 is $\Psi' = c_\phi \Psi$, up to an irrelevant sign, and $a_\phi \Psi' = \Psi$. Let $\mathcal{U}$ be an increasing measurable configuration event, and let A_n, A_{n+1} be its symmetric ordered sectors of cardinalities $n, n+1$. Put $f = \mathbf{1}_{A_n} \Psi$. The function $c_\phi f$ vanishes off A_{n+1} : any nonzero summand exhibits a subconfiguration belonging to $\mathcal{U}$. Hence, with $p_i = Q_i(\mathcal{U})$,

$$p_1 = \langle \Psi, f \rangle = \langle \Psi', c_\phi f \rangle = \langle \mathbf{1}_{A_{n+1}} \Psi', c_\phi f \rangle \leq \sqrt{p_2} \|c_\phi f\|_2 \leq \sqrt{p_2 p_1}.$$

The inner product is real and nonnegative because its value is p_1 ; Cauchy–Schwarz justifies the displayed inequality. Thus $p_1 \leq p_2$. Add an orthonormal basis of $H_2 \ominus H_1$ one vector at a time and then use the layer-cake formula to obtain the assertion for every bounded increasing measurable function.

For a measurable event F of the restriction to B , both functions

$$\eta \longmapsto \eta(B) + \mathbf{1}_F(\eta|_B), \quad \eta \longmapsto \eta(B) - \mathbf{1}_F(\eta|_B)$$

are increasing under inclusion. Indeed, an added point outside B changes neither term; a strict inclusion on B increases its count by at least one and changes the indicator by at most one. Counts are bounded by the finite ranks on the two laws; the increasing functions can therefore be truncated outside that range when applying the preceding assertion. Applying the assertion to both signs gives

$$|(Q_2|_B)(F) - (Q_1|_B)(F)| \leq \mathbb{E}_{Q_2} \Xi(B) - \mathbb{E}_{Q_1} \Xi(B).$$

Taking a supremum over F proves the total-variation inequality. $\square$

Proof of Theorem ??. Put $A = B_r$, $E = \mathbb{D} \setminus A$, and $B = \mathbb{D} \setminus B_R$. There are almost surely no particles on either bounding circle. The event $\{\Xi_N(A) = 0\}$ has positive probability, since the joint density is positive almost everywhere on E^N . Let Q_0 be the conditional law of the entire configuration in E given that event.

The Vandermonde determinant and Gram orthonormalization identify Q_0 as the projection determinantal process onto

$$H_0 = \text{Pol}_{N-1} \quad \text{in } L^2(E, \mu).$$

More generally, condition on the complete configuration $x = \{x_1, \dots, x_m\}$ in A . Almost surely its points are distinct, and $0 \leq m \leq N$. Factoring the joint Vandermonde shows that the conditional density of the $N - m$ exterior points is proportional to

$$|\Delta_{N-m}(z_1, \dots, z_{N-m})|^2 \prod_{i=1}^{N-m} |p_x(z_i)|^2 \prod_{i=1}^{N-m} d\mu(z_i), \quad p_x(z) = \prod_{a \in x} (z - a), \quad z_i \in E. \quad (21)$$

Terms depending only on x cancel in the conditional normalization. The density in (??) is the projection determinantal law onto

$$H_x = p_x \text{Pol}_{N-m-1} \subseteq H_0 \quad \text{in } L^2(E, \mu).$$

For $m = N$ the subspace is zero and the exterior is empty. Otherwise the identity follows by taking a determinant of any basis of H_x ; the factors p_x give exactly the displayed one-body

weight. Distinct interpolation nodes impose m independent conditions on Pol_{N-1} , so the displayed subspace also consists of all polynomials in H_0 vanishing at every point of x .

Write Q_x for this conditional exterior law and P_x for its projection kernel in $L^2(E, \mu)$. The subspace inclusion gives $P_x \leq P_0$. Applying Lemma ?? on B yields

$$d_{\text{TV}}(Q_x|_B, Q_0|_B) \leq \mathbb{E}_{Q_0} \Xi(B) - \mathbb{E}_{Q_x} \Xi(B). \quad (22)$$

No regularity or typicality requirement on the observed points x is needed for this inequality, beyond distinctness and admissibility.

Let $\bar{Q} = \text{Law}(\Xi_N|_E)$, the mixture of Q_x against the actual law of X_N . Convexity of total variation and (??) imply

$$d_{\text{TV}}(\bar{Q}|_B, Q_0|_B) \leq \mathbb{E} d_{\text{TV}}(Q_{X_N}|_B, Q_0|_B) \leq \Delta_N(r, R),$$

where

$$\Delta_N(r, R) = \mathbb{E}[\Xi_N(B) \mid \Xi_N(A) = 0] - \mathbb{E}\Xi_N(B).$$

A second triangle inequality and (??) give

$$\mathcal{B}(X_N, Y_{N,R}) \leq 2\Delta_N(r, R). \quad (23)$$

It remains to compute this scalar excess count. Angular orthogonality in E gives the orthonormal basis

$$\left\{ \sqrt{\frac{j}{1-r^{2j}}} z^{j-1} : 1 \leq j \leq N \right\}$$

of H_0 . Thus

$$\mathbb{E}[\Xi_N(B) \mid \Xi_N(A) = 0] = \sum_{j=1}^N \frac{1-R^{2j}}{1-r^{2j}}, \quad \mathbb{E}\Xi_N(B) = \sum_{j=1}^N (1-R^{2j}).$$

Subtracting proves the sum in (??). Finally $1-R^{2j} \leq j(1-R^2)$ and $1-r^{2j} \geq 1-r^2$. Summing $\sum_{j \geq 1} j r^{2j} = r^2/(1-r^2)^2$ proves (??). $\square$

Corollary 5.3 (Typical exteriors and normalized boundary changes). *Let $b_r(R)$ denote the last expression in (??). For every $\varepsilon > 0$,*

$$\mathbb{P}\{d_{\text{TV}}(\text{Law}(X_N \mid Y_{N,R}), \text{Law}(X_N)) > \varepsilon\} \leq \frac{b_r(R)}{\varepsilon}.$$

Suppose a changed probability law has density $W(Y_{N,R})$ relative to the actual finite gas, where $W \geq 0$ and $\mathbb{E}W = 1$. If $1 < p \leq \infty$ and $q = p/(p-1)$, then

$$d_{\text{TV}}(\text{Law}_W(X_N), \text{Law}(X_N)) \leq \|W\|_{L^p} b_r(R)^{1/q}. \quad (24)$$

For $p = \infty$, the convention is $q = 1$. Thus bounded normalized likelihoods, and more generally uniformly L^p normalized likelihoods, give a rigorously admissible class of boundary changes.

Proof. The first assertion is Markov's inequality. Set $d(Y) = d_{\text{TV}}(\text{Law}(X_N \mid Y), \text{Law}(X_N))$. For each local event F , conditioning on Y gives

$$|\mathbb{E}[W \mathbf{1}_F] - \mathbb{P}(F)| \leq \mathbb{E}[W d(Y)].$$

Take a supremum over F , use Hölder's inequality, and use $0 \leq d \leq 1$ and $\mathbb{E}d \leq b_r(R)$. $\square$

Remark 5.4 (What the benchmark excludes and what it does not). The estimate is uniform in N with r fixed. Consequently no measurable estimator built from the actual far exterior can preserve a positive amount of total-variation information about the whole fixed local configuration along $R \uparrow 1$, even if N changes arbitrarily. This includes exterior harmonic sums, exterior counts, and functions whose formula depends on N and R . It is stronger than invoking tail triviality of the limiting Bergman process after taking N to infinity first.

The constant is not uniform as the observation radius r tends to one. Nor is the conditional distance small for every individual exterior configuration: the theorem gives an average and hence a high-probability bound under the selected boundary law. A rare boundary event can have a normalized likelihood with an arbitrarily large L^p norm in (??). Therefore the result is consistent with adverse deterministic exterior configurations.

5.1 A formulation transferable beyond determinants

The reversal of conditioning in the preceding proof is a probability argument, not a determinantal one. It gives a concrete alternative target for screening at other temperatures.

Proposition 5.5 (Full-law attenuation from a hole response). *Let Ξ be any finite point process on a standard Borel region D , and let $A, B \subset D$ be disjoint measurable sets. Assume $\mathbb{P}\{\Xi(A) = 0\} > 0$ and that all the expectations below are finite. Write Q_0 for the law of $\Xi|_{D \setminus A}$ conditioned on the hole event and Q_x for its law conditioned on the complete configuration $\Xi|_A = x$. Suppose that, for almost every x under the actual interior marginal, $Q_x|_B$ is stochastically dominated by $Q_0|_B$ under inclusion; an inclusion coupling is a sufficient condition. Then*

$$\mathcal{B}(\Xi|_A, \Xi|_B) \leq 2 \{ \mathbb{E}[\Xi(B) \mid \Xi(A) = 0] - \mathbb{E}\Xi(B) \}. \quad (25)$$

More generally, it suffices that for some fixed reference law Q_* and measurable $e(x) \geq 0$,

$$d_{\text{TV}}(Q_x|_B, Q_*|_B) \leq e(x), \quad \mathbb{E}e(\Xi|_A) < \infty.$$

Then $\mathcal{B}(\Xi|_A, \Xi|_B) \leq 2\mathbb{E}e(\Xi|_A)$.

Proof. Apply stochastic domination to the two increasing functions $\eta(B) \pm \mathbf{1}_F(\eta|_B)$, truncated if necessary and then passed to the limit by the assumed finite count expectations. As in the proof of Lemma ??, this bounds the distance to $Q_0|_B$ by its excess expected count. Integrate over the actual interior configuration. Convexity bounds the distance between the unconditional exterior and the same reference law by that integral. The triangle inequality and (??) prove both assertions. $\square$

For the actual disk gas this proposes two separate non-determinantal estimates: an inclusion coupling, or a controlled approximate replacement, for complete interior conditioning; and vanishing far-exterior excess count after imposing a fixed inner hole. The susceptibility inequality at $0 < \beta \leq 2$ alone does not establish either assertion. In particular, a two-particle-to-one-particle deletion coupling cannot be iterated over arbitrary particle numbers without a further theorem. Proposition ?? is thus a proved reduction, not a claim of full screening below two.

6 Why susceptibility alone does not give deletion couplings

The susceptibility estimate does not, by itself, justify replacing an exterior conditional law by a configuration containing all its particles. The following example isolates the logical gap. It is an example of a discrete probability measure, not a counterexample for the disk gas.

Proposition 6.1 (A Lorentzian measure without deletion domination). *There is a strictly positive probability measure on every three-element subset of a five-element set whose generating polynomial is Lorentzian, but which fails stochastic covering after conditioning on one coordinate.*

Proof. Use coordinates x_1, x_2, x_3, y, z , and put

$$F = e_3(x_1, x_2, x_3, y, z) + 3x_1x_2x_3,$$

where e_3 is the sum of all squarefree monomials of degree three. Thus the coefficient of $x_1x_2x_3$ is four and each of the other nine coefficients is one. The support is the set of all three-element subsets, which has the matroid exchange property.

For a homogeneous cubic with this support, the Lorentzian criterion used above asks that each first derivative have a Hessian with at most one positive eigenvalue. The derivatives with respect to y and z are e_2 on the four remaining variables. Their nonzero Hessian block is the matrix with zero diagonal and all off-diagonal entries one, whose eigenvalues are $3, -1, -1, -1$.

For the derivative with respect to x_1 , order the remaining variables as x_2, x_3, y, z . Its Hessian block is

$$H = \begin{pmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

The vectors $(1, -1, 0, 0)$ and $(0, 0, 1, -1)$ have eigenvalues -4 and -1 . On vectors (a, a, b, b) the action is represented by $\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$, whose eigenvalues are 5 and 0 . The derivatives with respect to x_2 and x_3 have the same spectrum. The unused differentiated coordinate adds only a zero eigenvalue. This verifies the complete Lorentzian criterion.

Normalize the coefficients of F to a probability measure μ . Since $F(1, 1, 1, 1, 1) = 13$, and writing I_y, I_z for occupancy indicators,

$$\mu(I_y = 1) = \mu(I_z = 1) = \frac{6}{13}, \quad \mu(I_y = I_z = 1) = \frac{3}{13}.$$

Consequently

$$\text{Cov}_\mu(I_y, I_z) = \frac{3}{13} - \frac{36}{169} = \frac{3}{169} > 0.$$

More explicitly, the law on the other four coordinates given $I_y = 1$ has two particles and assigns probability $1/2$ to $I_z = 1$; the law given $I_y = 0$ has three particles and assigns probability $3/7$ to that same increasing event. The two-particle configuration therefore cannot be a subset of the three-particle configuration under any coupling. $\square$

Remark 6.2. This example is not asserted to have coefficients of the form $\prod_{\{i,j\} \subset S} |a_i - a_j|^\beta$ for planar points a_i . It disproves the implication from the already established Lorentzian property to deletion domination. Additional structure of the disk interaction could still imply the latter.

For precision, the additional property has a finite system of exact inequalities. Fix $\beta > 0$ and $1 \leq n \leq q$. Let $a_1, \dots, a_q$ be distinct points, let y be a further point, and let $w_i > 0$. For a subset $T \subset \{1, \dots, q\}$ of size n and a subset S of size $n-1$, set

$$A_T = \prod_{i \in T} w_i \prod_{\{i,j\} \subset T} |a_i - a_j|^\beta, \quad B_S = \prod_{i \in S} (w_i |a_i - y|^\beta) \prod_{\{i,j\} \subset S} |a_i - a_j|^\beta.$$

Write $Z_A = \sum_{|T|=n} A_T$ and $Z_B = \sum_{|S|=n-1} B_S$. For a family $\mathcal{C}$ of $(n-1)$ -element subsets, define its up-shadow by

$$\partial^+ \mathcal{C} = \{T : |T| = n, S \subset T \text{ for some } S \in \mathcal{C}\}.$$

Proposition 6.3 (The exact finite deletion criterion). *A coupling of $Q(S) = B_S/Z_B$ and $P(T) = A_T/Z_A$ supported on $S \subset T$ exists if and only if, for every family $\mathcal{C}$,*

$$Z_A \sum_{S \in \mathcal{C}} B_S \leq Z_B \sum_{T \in \partial^+ \mathcal{C}} A_T. \tag{26}$$

Proof. Necessity follows because $S \in \mathcal{C}$ and $S \subset T$ imply $T \in \partial^+ \mathcal{C}$. For sufficiency, build a finite network with a source, one vertex for each S , one vertex for each T , and a sink. Give source-to- S edges capacity $Q(S)$, T -to-sink edges capacity $P(T)$, and each inclusion edge $S \rightarrow T$ capacity two. A cut of capacity less than one cannot cross an inclusion edge. If its source side contains a family $\mathcal{C}$ of lower vertices, it must therefore contain their upper shadow, and its capacity is at least $1 - Q(\mathcal{C}) + P(\partial^+ \mathcal{C}) \geq 1$. All other cuts also have capacity at least one. The max-flow/min-cut theorem gives a flow of value one. Since both sets of marginal capacities sum to one, all source and sink edges are saturated, and the flows on inclusion edges form the required coupling. $\square$

For $n = 2$, the families $\mathcal{C}$ are just sets of possible singleton locations; the Hilbert-space calculation in the two-to-one deletion theorem proves all these inequalities. For $n \geq 3$, families of multi-particle configurations must be considered. Linear-statistic variance bounds do not check those families. Establishing (??) for all relevant planar distance weights, or a suitably restricted version for boundary data selected by the actual gas, remains an explicit possible route to a count-sensitive comparison.

7 An exact finite-range splitting and a uniformly controlled reference gas

This section concerns the actual unweighted disk gas. It supplies an exact representation, including every particle-number weight. The representation does not by itself prove screening: its final, positive quadratic tilt still has to be controlled. All measures and kernels used below are constructed explicitly.

Put $\mu = \pi^{-1} dA$, $\nu = \pi^{-1}(1 - |z|^2)^{-2} dA$, and let d denote hyperbolic distance for curvature -1 . Write

$$C(z, w) = -\log |z - w| = g(d(z, w)) + h(z, w),$$

$$g(r) = -\log \tanh(r/2), \quad h(z, w) = -\log |1 - z\bar{w}|.$$

The normalized radial element of ν about any point is $\frac{1}{2} \sinh r \, dr$. For $a > 0$, let S_a average a function over the hyperbolic sphere of radius a , with its rotational probability measure. This operator preserves harmonic functions. It preserves ν and is self-adjoint on $L^2(\nu)$: the invariant measure on pairs of points at distance a is unchanged on exchanging the points. Source and target averaging of an invariant radial kernel commute.

Proposition 7.1 (Exact finite-range decomposition). *Let C_a be C averaged by S_a in each argument. There is a nonnegative radial kernel k_a such that*

$$C(z, w) = C_a(z, w) + k_a(d(z, w)), \quad k_a(r) = 0 \quad (r \geq 2a).$$

More precisely, with $q_a(r) = (g(r) - g(a))_+$,

$$k_a = q_a + S_a q_a, \tag{27}$$

$$M_a := \int k_a(d(o, z)) \nu(dz) = 2 \log \cosh(a/2), \tag{28}$$

$$C_a(z, z) = g(a) - \log(1 - |z|^2). \tag{29}$$

The kernel C_a is finite and positive semidefinite. For all $\beta > 0$ and $\alpha \geq 0$,

$$\int e^{\alpha d(o, z)} (1 - e^{-\beta k_a(d(o, z))}) \nu(dz) \leq \beta e^{2\alpha a} M_a. \tag{30}$$

Proof. The spherical mean of the Green potential is

$$S_a g(r) = g(\max\{r, a\}).$$

Outside the sphere this is the harmonic mean-value property. Inside, the sphere potential is a rotationally invariant harmonic function regular at its center, hence constant; its value at the center is $g(a)$. These assertions agree continuously on the sphere, where the logarithmic singularity in the angular integral is integrable. Since h is harmonic in each argument, its spherical means equal h . Therefore $C_a = S_a^2 g + h$ and $g - S_a^2 g = (g - S_a g) + S_a(g - S_a g) = q_a + S_a q_a$. This proves positivity and the range statement by the triangle inequality. Using $g'(r) = -1/\sinh r$ and integration by parts gives

$$\int q_a d\nu = \frac{1}{2} \int_0^a (g(r) - g(a)) \sinh r dr = \frac{1}{2} \int_0^a \frac{\cosh r - 1}{\sinh r} dr = \log \cosh(a/2).$$

Measure preservation by S_a gives the stated value of M_a . Furthermore

$$S_a^2 g(0) = S_a[g(\max\{r, a\})](0) = g(a),$$

which proves the diagonal formula.

For completeness, positivity as a quadratic form is a different assertion from pointwise positivity of k_a . For any real finite signed measure σ of finite Green energy, $\iint g d\sigma d\sigma \geq 0$. One way to verify this is to first use a smooth charge density compactly supported in a disk B_R : its zero-boundary Green potential U satisfies $-\Delta U = 2\pi\sigma$, and integration by parts gives the energy $(2\pi)^{-1} \int_{B_R} |\nabla U|^2 dA \geq 0$. Increasing R to 1 gives the full-disk assertion; approximating the charges gives it for finite-energy measures. In the present application the charges are finite linear combinations of uniform circle measures, whose logarithmic energies are finite. Their energies converge under ordinary smooth approximation, because the only local singularity is logarithmic. Thus $S_a^2 g$ is positive semidefinite on point configurations. Also

$$\sum_{i,j} t_i t_j h(z_i, z_j) = \sum_{k \geq 1} \frac{1}{k} \left| \sum_i t_i z_i^k \right|^2 \geq 0 \quad (t_i \in \mathbb{R}).$$

The series converges for every finite configuration in $\mathbb{D}$. This proves the assertion for C_a . Finally, $1 - e^{-\beta k_a} \leq \beta k_a$ and its support is contained in B_{2a} , proving (??). $\square$

For $\eta = \sum_{i=1}^N \delta_{z_i}$ at distinct points, the proposition gives the following identity without any approximation:

$$|\Delta_N(\mathbf{z})|^\beta \mu^N(d\mathbf{z}) = e^{\beta N g(a)/2} e^{-\frac{\beta}{2} C_a(\eta, \eta)} e^{-\beta \sum_{i < j} k_a(d(z_i, z_j))} \prod_i (1 - |z_i|^2)^{2-\beta/2} \nu(dz_i). \quad (31)$$

Here $C_a(\sigma, \tau) = \iint C_a(z, w) \sigma(dz) \tau(dw)$ whenever the integral is defined. The one-body factor is the retained diagonal self-energy, together with the change of area measure. It cannot be dropped when comparing particle counts.

7.1 Construction of the reference density

We now assume $0 < \beta < 2$, and set

$$p = 2 - \beta/2 > 1, \quad c = \frac{2p}{\beta} = \frac{4}{\beta} - 1 > 0, \quad u(z) = (1 - |z|^2)^p.$$

The appearance of c below concerns a reference density, not the unproved intensity of the actual particle process.

Proposition 7.2 (A finite-mass reference at every activity). *For every $\lambda > 0$ there exists a positive radial smooth function b_λ on $\mathbb{D}$ satisfying*

$$b_\lambda(z) = \lambda u(z) \exp[-\beta(C * b_\lambda)(z)], \quad (C * b)(z) = \int C(z, w)b(w)\nu(dw). \quad (32)$$

It has finite mass, is radially decreasing, and satisfies

$$0 < b_\lambda(z) < c, \quad b_\lambda(z) \sim \lambda(1 - |z|^2)^p \quad (|z| \uparrow 1).$$

These references increase strictly with λ and

$$b_\lambda \longrightarrow c \quad \text{locally uniformly as } \lambda \longrightarrow \infty. \quad (33)$$

Their defining local equation is

$$\Delta_{\mathbb{H}} \log b_\lambda = \frac{\beta}{2}(b_\lambda - c). \quad (34)$$

Proof. We construct the functions by a radial ordinary differential equation, which also proves the required bounds without an assumption about the gas. Let $t = d(0, z)$, and fix $x \in (0, c)$. Solve

$$v''(t) + \coth t v'(t) = \frac{\beta}{2}(e^{v(t)} - c), \quad v(0) = \log x, \quad v'(0) = 0. \quad (35)$$

The regular radial solution exists uniquely near zero. For example, the integrated equation

$$v(t) = \log x + \frac{\beta}{2} \int_0^t \frac{1}{\sinh s} \int_0^s (e^{v(r)} - c) \sinh r \, dr \, ds$$

has a unique continuous solution on a short interval by the contraction mapping theorem, since its double-integral operator has norm $O(t^2)$ on bounded sets. It then gives the usual regularity and the differential equation. As long as the solution exists, the same formula shows that $v'(t) < 0$ for $t > 0$ and $e^{v(t)} \leq x < c$: a first failure of either property is impossible. Moreover

$$-p \tanh(t/2) \leq v'(t) \leq -\frac{\beta}{2}(c - x) \tanh(t/2).$$

These bounds prevent finite-time blowup and extend the solution to all $t \geq 0$. If $b_x(t) = e^{v(t)}$ and $\delta_x = \beta(c - x)/2 > 0$, they imply

$$x \cosh(t/2)^{-2p} \leq b_x(t) \leq x \cosh(t/2)^{-2\delta_x}. \quad (36)$$

An exact refinement of the integrated equation is

$$v'(t) = -p \tanh(t/2) + \frac{\beta}{2 \sinh t} \int_0^t b_x(s) \sinh s \, ds.$$

The integral of its second term over $t \in (0, \infty)$ is finite. Indeed, Tonelli identifies it with

$$\frac{\beta}{2} \int_0^\infty b_x(s) \sinh s \left(\int_s^\infty \frac{dt}{\sinh t} \right) ds = \frac{\beta}{2} \int_0^\infty b_x(s) \sinh s g(s) \, ds < \infty.$$

Near zero the integrand is $O(s|\log s|)$; at infinity, $\sinh s g(s)$ is bounded and (??) gives exponential decay. Consequently

$$b_x(t) \sim \lambda(x) \cosh(t/2)^{-2p}, \quad (37)$$

$$\lambda(x) = x \exp \left[\frac{\beta}{2} \int_0^\infty b_x(s) \sinh s g(s) \, ds \right]. \quad (38)$$

Since $p > 1$, (??) proves that $b_x \nu$ has finite total mass.

Let $B_x = C * b_x$. Angular averaging gives $C(z, w) \mapsto -\log \max\{|z|, |w|\}$ for radial charges. In particular $B_x \geq 0$, it is bounded by $B_x(0) < \infty$, and $B_x(z) \rightarrow 0$ as $|z| \uparrow 1$. The latter also follows from $B_x(r) \leq -\log r \int b_x d\nu$ for $r > 0$. Its Laplacian is $\Delta_{\mathbb{H}} B_x = -b_x/2$, since $\Delta C(\cdot, w) = -2\pi\delta_w$ in Euclidean coordinates. Also $\Delta_{\mathbb{H}} \log(1 - |z|^2) = -1$. It follows from (??) that $\log b_x - p \log(1 - |z|^2) + \beta B_x$ is a radial harmonic function regular at zero and hence constant. Taking its boundary limit using (??) shows that this constant is $\log \lambda(x)$. Thus (??) holds for b_x at activity $\lambda(x)$.

It remains to parameterize all activities and prove the limiting claim. If $0 < x < y < c$, the difference $w = v_y - v_x$ starts positive with zero derivative. While it is positive, $(\sinh t w')' = \frac{\beta}{2} \sinh t (b_y - b_x) > 0$ for $t > 0$. It therefore remains positive and increasing. Thus $b_y > b_x$ and, by the formula for λ , $\lambda(y) > \lambda(x)$. Continuity of $\lambda(x)$ inside $(0, c)$ follows from continuous dependence of the ODE on initial values and dominated convergence in the integral formula, using the upper bound in (??) uniformly on compact x -intervals. As $x \downarrow 0$, the same bound, with for example $\delta_x \geq p/2$, shows that the integral in $\log(\lambda(x)/x)$ tends to zero; hence $\lambda(x) \rightarrow 0$. As $x \uparrow c$, the ODE solutions tend on compact intervals to the constant solution $b = c$. Monotonicity and Tonelli give

$$\int_0^\infty b_x(s) \sinh s g(s) ds \rightarrow \infty,$$

because $\sinh s g(s) \rightarrow 1$. Therefore $\lambda(x) \rightarrow \infty$. The continuous strictly increasing map $x \mapsto \lambda(x)$ maps $(0, c)$ onto $(0, \infty)$. Its inverse defines b_λ and proves all the claims, including local uniform convergence by the continuous dependence already used. $\square$

7.2 Exact grandcanonical representation and the small reference norm

Let $P_{\zeta, \beta}$ denote the grandcanonical mixture of the original unweighted gases, with unnormalized N -particle sector

$$\frac{\zeta^N}{N!} |\Delta_N(\mathbf{z})|^\beta \prod_i \mu(dz_i).$$

For $0 < \beta < 2$ its partition function is finite. Indeed, $\int |\Delta_N|^2 d\mu^N = N! \prod_{j=0}^{N-1} (j+1)^{-1} = 1$ by expanding the determinant and using angular orthogonality. Jensen's inequality gives $Z_N(\beta) \leq 1$, and hence $\sum_N \zeta^N Z_N(\beta)/N! \leq e^\zeta$.

Theorem 7.3 (The actual gas as a tilt of a uniformly subcritical reference). *Fix $a > 0$, $\zeta > 0$, and $0 < \beta < 2$. Set*

$$\lambda = \zeta e^{\beta g(a)/2}, \quad b = b_\lambda, \quad a_\lambda(z) = b(z) \exp[\beta(k_a * b)(z)].$$

Let Q_λ be the finite grandcanonical gas with activity $a_\lambda(z)\nu(dz)$ and pair interaction βk_a . Then

$$\frac{dP_{\zeta, \beta}}{dQ_\lambda}(\eta) = \frac{\exp[-\frac{\beta}{2} C_a(\eta - b\nu, \eta - b\nu)]}{E_{Q_\lambda} \exp[-\frac{\beta}{2} C_a(\eta - b\nu, \eta - b\nu)]}. \quad (39)$$

The expectation in the denominator is strictly positive. Uniformly in λ ,

$$a_\lambda(z) \leq c \exp(\beta c M_a), \quad (40)$$

$$\sup_z \int e^{\alpha d(z, w)} a_\lambda(w) (1 - e^{-\beta k_a(d(z, w))}) \nu(dw) \leq \beta c M_a \exp(\beta c M_a + 2\alpha a). \quad (41)$$

In particular, if $0 < \beta \leq 1/4$ and $a = \sqrt{\beta}$, the right-hand side for $\alpha = 1$ is at most $\frac{1}{4} e^{5/4} < 1$.

Proof. The convolution $k_a * b$ is nonnegative and at most cM_a , so $a_\lambda \leq e^{\beta c M_a} b$. Its integral is finite by Proposition ???. Since the pair interaction is nonnegative, the grand partition function of Q_λ lies between 1 and $\exp(\int a_\lambda d\nu)$, and Q_λ has a finite number of particles almost surely.

All energies in (??) are finite. Indeed $C * b$ is bounded and $b\nu$ has finite mass. Radial symmetry annihilates its nonconstant harmonic moments, so $h * b = 0$ and $C_a * b = S_a^2 g * b$, which lies between 0 and $C * b$ because $0 \leq S_a^2 g \leq g$. This controls the deterministic and cross energies; the finite atomic self-energies are covered by Proposition ??. Moreover $\int [-\log(1 - |z|^2)] b(z) \nu(dz) < \infty$ by the boundary asymptotic $b \sim \lambda(1 - |z|^2)^p$ with $p > 1$. Together with $|h(z, w)| \leq \frac{1}{2}(h(z, z) + h(w, w))$, which follows from the positive semidefiniteness of h and the arithmetic–geometric mean inequality, this justifies absolute integration of its harmonic term. Truncating the radial background and then passing to the limit consequently extends the positive-energy formula to this noncompact finite measure. The full energy of the signed charge is nonnegative by the same Green-energy argument and harmonic moment expansion. Hence the tilt lies in $(0, 1]$ on every finite configuration and has positive expectation.

To prove the identity, apply (??) to every N -sector and use $\log(\lambda u) = \log b + \beta C * b$ and $C = C_a + k_a$. Completing the square gives the unnormalized equality

$$\begin{aligned} & e^{-\frac{\beta}{2} C_a(\eta, \eta)} e^{-\beta \sum_{i < j} k_a(d(z_i, z_j))} \prod_i \lambda u(z_i) \nu(dz_i) \\ &= e^{\frac{\beta}{2} C_a(b\nu, b\nu)} e^{-\frac{\beta}{2} C_a(\eta - b\nu, \eta - b\nu)} e^{-\beta \sum_{i < j} k_a(d(z_i, z_j))} \prod_i a_\lambda(z_i) \nu(dz_i). \end{aligned}$$

This also holds for $N = 0$; the two deterministic energy factors then cancel. Division by $N!$ and summation over N prove (??), including its particle-count weights.

The first bound follows from $b < c$; the second follows from (??). Finally, $\log \cosh x \leq x^2/2$ gives

$$M_a = 2 \log \cosh(a/2) \leq a^2/4.$$

For $a = \sqrt{\beta}$, $\beta c M_a \leq (4 - \beta)\beta/4 \leq \beta$, so the right-hand side of (??) is at most $\beta e^{\beta+2\sqrt{\beta}} \leq \frac{1}{4} e^{5/4} < 1$. $\square$

Corollary 7.4 (A canonical reference at every particle number). *For every $N \geq 1$ there is a unique $\lambda_N > 0$ such that $\int b_{\lambda_N} d\nu = N$. Moreover $\lambda_N \rightarrow \infty$ and $b_{\lambda_N} \rightarrow c$ locally uniformly. Let Q_N be the canonical N -particle gas with one-body weight $a_{\lambda_N} \nu$ and pair interaction βk_a . Then the original unweighted canonical gas satisfies exactly*

$$\frac{dP_{N,\beta}}{dQ_N}(\eta) = \frac{e^{-\frac{\beta}{2} C_a(\eta - b_{\lambda_N} \nu, \eta - b_{\lambda_N} \nu)}}{E_{Q_N} e^{-\frac{\beta}{2} C_a(\eta - b_{\lambda_N} \nu, \eta - b_{\lambda_N} \nu)}}.$$

Proof. The angular formula in the proof of Proposition ?? gives $C * b_\lambda \geq 0$. Hence (??) implies $b_\lambda \leq \lambda u$, where $\int u d\nu = (p - 1)^{-1} < \infty$. The total mass is consequently continuous in λ by dominated convergence on compact activity intervals, strictly increasing, and tends to zero as $\lambda \downarrow 0$. By monotone convergence and (??), it tends to $c\nu(\mathbb{D}) = \infty$ as $\lambda \rightarrow \infty$. This proves the assertions about λ_N and the reference limit. In Theorem ??, choose $\zeta_N = \lambda_N e^{-\beta g(a)/2}$ and condition both sides on total count N . Every such sector has positive probability. The left conditional law is exactly $P_{N,\beta}$, and the right reference law is Q_N , giving the displayed identity. $\square$

Remark 7.5 (What the small norm establishes, and what it does not). The positive kernel $T(z, dw) = a_\lambda(w)(1 - e^{-\beta k_a(d(z, w))})\nu(dw)$ has a weighted row sum strictly below one in the stated range. Consequently its repeated path integrals are summable. More explicitly, if the right side of (??) is $q < 1$, then the total weight of paths starting at z , of any positive length, whose endpoint is at distance at least L from z , is bounded by $e^{-\alpha L} q / (1 - q)$. To verify this,

insert the weight $e^\alpha \sum d(z_{i-1}, z_i)$, use the triangle inequality, and integrate successive vertices; an n -step path has weighted integral at most q^n . This includes hyperbolic volume growth exactly in the row integrals. It is the quantitative smallness required by finite-range comparison and disagreement constructions for the reference gas.

Equation (??) still contains the smooth, long-range quadratic energy. A nonnegative energy can create dependencies after normalization; its positivity alone does not allow the reference path bound to be transferred to the tilted law. A uniform estimate on this tilt under local changes, including its normalizing ratios, remains unproved. Furthermore $b_\lambda \rightarrow c$ is a theorem about the explicitly constructed reference; it is not an identification of the intensity of $P_{\zeta, \beta}$ or of its fixed- N subsequences. One can work directly with the canonical identity in Corollary ??, but conditioning the reference on its total count introduces an additional global dependency which the grandcanonical row-norm bound alone does not control. Alternatively, passing a result for the grandcanonical mixture back to every N requires a comparison of nearby particle numbers. The gain here is an exact starting point for resummation with a proved activity bound and a small finite-range remainder, rather than an assumed background density or a replacement of the original interaction by a massive one.

7.3 An exact restoring potential in the imaginary-field representation

The short-range expansion can be controlled uniformly even at complex activities. This provides a further rigorous step in a Debye construction. The following elementary version includes its proof to specify which uniformity is obtained.

Proposition 7.6 (Complex-activity coercivity for a repulsive gas). *Let $A(x)m(dx)$ be a finite nonnegative measure, let $v(x, y) \geq 0$ be a measurable symmetric pair interaction, and set*

$$p(x, y) = 1 - e^{-v(x, y)}, \quad q = \sup_x \int A(y)p(x, y)m(dy).$$

Define the complex partition function

$$\mathcal{Z}_\phi = \sum_{n \geq 0} \frac{1}{n!} \int \prod_i A(x_i) e^{i\phi(x_i)} \prod_{i < j} e^{-v(x_i, x_j)} dm^n.$$

If $q \leq 1/19$, then $\mathcal{Z}_\phi \neq 0$ for every real measurable ϕ , and

$$\log |\mathcal{Z}_\phi| - \log \mathcal{Z}_0 \leq -\frac{1}{2} \int A(x)(1 - \cos \phi(x))m(dx). \quad (42)$$

Proof. Put $f = -p$. For $n \geq 2$ let $U_n(\mathbf{x}) = \sum_{G \text{ connected}} \prod_{\{i, j\} \in E(G)} f(x_i, x_j)$, and put $U_1 = 1$. The repulsive tree-graph bound is

$$|U_n(\mathbf{x})| \leq \sum_{T \text{ tree}} \prod_{\{i, j\} \in E(T)} p(x_i, x_j). \quad (43)$$

Here is a direct verification. Root a connected labelled graph at vertex 1, form its distance levels, and give each other vertex the smallest labelled parent among its neighbors in the preceding level. This maps the graph to a rooted tree T . The graphs giving T are precisely $T \subset G \subset R(T)$, where $R(T)$ permits all edges within a level and all edges to the preceding level whose endpoint there is at least the chosen parent's label. Summing over this interval of graphs gives

$$\prod_{e \in T} f_e \prod_{e \in R(T) \setminus T} (1 + f_e).$$

Since $0 \leq 1 + f_e \leq 1$, summing these disjoint intervals proves (??).

Fixing a root, successive integration of leaves bounds a tree integral by q^{n-1} times the root integral. There are n^{n-2} labelled trees, as counted by their Prüfer codes. Consequently the connected series

$$L_\phi = \sum_{n \geq 1} \frac{1}{n!} \int U_n(\mathbf{x}) \prod_i A(x_i) e^{i\phi(x_i)} dm^n \quad (44)$$

converges absolutely when $eq < 1$, uniformly in the phase. More generally it converges when the activity is multiplied by a complex scalar t with $|t|eq < 1$. The identity $\exp L_\phi = \mathcal{Z}_\phi$ first holds as a power-series identity near $t = 0$: each graph decomposes uniquely into connected components, which is precisely the exponential formula for the partition series. Analytic continuation within $|t|eq < 1$ proves it at $t = 1$. This also proves nonvanishing and $\operatorname{Re} L_\phi = \log |\mathcal{Z}_\phi|$.

For any real numbers $\theta_1, \dots, \theta_n$,

$$1 - \cos(\theta_1 + \dots + \theta_n) \leq n \sum_{j=1}^n (1 - \cos \theta_j).$$

Indeed, use the telescoping bound $|\prod_j e^{i\theta_j} - 1| \leq \sum_j |e^{i\theta_j} - 1|$ and Cauchy-Schwarz, then divide the squared inequality by 2. The $n = 1$ term in $\operatorname{Re}(L_\phi - L_0)$ is $-I$, where $I = \int A(1 - \cos \phi) dm$. The preceding inequality, the tree bound and rooted leaf integration show that the sum of the absolute values of all remaining terms is at most $S(q)I$, with

$$S(q) = \sum_{n \geq 2} \frac{n^n}{n!} q^{n-1} \leq \frac{e^2 q}{1 - eq}.$$

The last bound uses $n! \geq (n/e)^n$. For $q \leq 1/19$, the elementary bound $e < 11/4$ gives $S(q) \leq 121/260 < 1/2$. Thus $\operatorname{Re}(L_\phi - L_0) \leq -(1 - S(q))I \leq -I/2$, proving (??). $\square$

Apply this proposition to the reference activity $A = a_\lambda$, measure $m = \nu$, and pair interaction $v = \beta k_a$. If $0 < \beta \leq 1/20$ and $a = \sqrt{\beta}$, then $q \leq \beta e^\beta \leq 1/19$, using $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$. All bounds are uniform in λ , including the choices $\lambda = \lambda_N$. This statement concerns the grandcanonical partition functional at those activities, not the partition function conditioned on count N . Indeed, a constant phase ϕ_0 multiplies a canonical partition function by $e^{iN\phi_0}$ and leaves its modulus unchanged. Thus the canonical constant phase is an explicit noncontracting mode.

There is an exact Gaussian formula behind this estimate. Let Y_a be a centered real Gaussian field with covariance C_a . It can be constructed by sphere-averaging a zero-boundary Gaussian free field and adding the independent harmonic Gaussian series with covariance h . The averaged field has a measurable version, finite at every interior point; its local mean-square continuity also follows from the integrable logarithmic circle averages defining C_a . The random variable $b(Y_a) = \int b(z)Y_a(z)\nu(dz)$ is well defined in L^2 , since $C_a(b\nu, b\nu) < \infty$. If Ξ_{short} denotes the unnormalized grand partition functional of the finite-range pair interaction, then

$$\Xi_{\zeta, \beta} = e^{\frac{\beta}{2} C_a(b\nu, b\nu)} E \left[e^{-i\sqrt{\beta} b(Y_a)} \Xi_{\text{short}}(a_\lambda e^{i\sqrt{\beta} Y_a}) \right]. \quad (45)$$

To verify it, expand the finite grand partition series and take the Gaussian characteristic function of $\sqrt{\beta}(\eta - b\nu)(Y_a)$. This gives exactly $e^{-\frac{\beta}{2} C_a(\eta - b\nu, \eta - b\nu)}$, and the identity follows from the square completion above. Fubini is justified because the absolute value before Gaussian averaging is bounded by the finite positive partition function $\Xi_{\text{short}}(a_\lambda)$.

For $\beta \leq 1/20$, Proposition ?? proves the pointwise field bound

$$\left| e^{-i\sqrt{\beta} b(Y_a)} \Xi_{\text{short}}(a_\lambda e^{i\sqrt{\beta} Y_a}) \right| \leq \Xi_{\text{short}}(a_\lambda) \exp \left[-\frac{1}{2} \int a_\lambda(z) (1 - \cos(\sqrt{\beta} Y_a(z))) \nu(dz) \right]. \quad (46)$$

On the part of the disk where $|\sqrt{\beta}Y_a| \leq \pi$, the integrand in the exponent is at least $\beta a_\lambda Y_a^2/\pi^2$, since $1 - \cos t \geq 2t^2/\pi^2$ for $|t| \leq \pi$. Thus the exact resummation has a quadratic restoring term in its small field region. It is periodic in the field and has additional wells. There is also an imaginary linear residue: the derivative of $\log \Xi_{\text{short}}(a_\lambda e^{i\sqrt{\beta}Y})$ at $Y = 0$ is $i\sqrt{\beta} \int \rho_{Q_\lambda} Y d\nu$. After multiplication by the centering phase in (??), it is $i\sqrt{\beta} \int (\rho_{Q_\lambda} - b)Y d\nu$. The constructed reference profile b is not asserted to equal ρ_{Q_λ} , so this term has not been canceled. Controlling these wells, the boundary field and ratios of the complex integrals in (??) remains necessary. An upper bound on the absolute integrand does not give a lower bound on its oscillatory expectation or a comparison of normalized local laws. These are the remaining large-field and normalization steps; they are not proved by the small reference norm or by (??).

8 The reference-density residue and an exactly centered reference

The reference profile b_λ in Proposition ?? is a deterministic Poisson–Boltzmann profile. It is not the intensity of the finite-range reference gas. We first bound their difference, and then construct a different reference whose centering is exactly its own intensity. These assertions concern the reference representation; neither identifies the intensity of the original disk gas.

Throughout this section, $0 < \beta \leq 1/20$, $a = \sqrt{\beta}$, $c = 4/\beta - 1$, $p = 2 - \beta/2$, $u(z) = (1 - |z|^2)^p$, $k = k_a$, and $M = M_a = 2 \log \cosh(a/2)$. We use the abbreviation

$$t = \beta c M \leq \beta(1 - \beta/4) \leq \frac{79}{1600} < \frac{1}{20}.$$

For a finite nonnegative activity $A\nu$, let Q_A denote the grandcanonical gas with this activity and pair interaction βk , and let ρ_A be its intensity relative to ν .

Lemma 8.1 (Elementary insertion identities). *For every such activity,*

$$\rho_A(z) = A(z) E_{Q_A} \exp \left[-\beta \sum_{w \in \eta} k(d(z, w)) \right], \quad (47)$$

$$\rho_A(z) \leq A(z), \quad \rho_A^{(2)}(z, w) \leq A(z)A(w). \quad (48)$$

*If $(k * \rho_A)(z)$ is finite, then also*

$$\rho_A(z) \geq A(z) \exp[-\beta(k * \rho_A)(z)]. \quad (49)$$

Proof. Expand the finite partition series and distinguish one of its labelled points. Reindexing the remaining count proves (??). Distinguishing two points gives the analogous formula with the extra factor $e^{-\beta k(d(z, w))}$ and both insertion factors. All these factors lie in $[0, 1]$, which proves (??). Tonelli applies because the series are nonnegative and bounded by the Poisson partition series. Finally Jensen’s inequality applied to the exponential in (??) proves (??). $\square$

Proposition 8.2 (A uniform second-order density residue). *Let $b = b_\lambda$ and $A = b \exp(\beta k * b)$ be the reference activity in Theorem ??. Then, uniformly in $\lambda > 0$ and $z \in \mathbb{D}$,*

$$|\rho_A(z) - b(z)| \leq 2\beta^2 b(z). \quad (50)$$

Proof. Write $s(z) = \beta(k * b)(z)$, so that $0 \leq s \leq t \leq \beta$ and $A \leq e^t b$. By the insertion identities and Jensen,

$$\rho_A \geq A e^{-\beta k * A} \geq b \exp[-(e^t - 1)s] \geq b e^{-d}, \quad d = t(e^t - 1) \leq \beta^2 e^\beta.$$

Since $e^\beta < 2$, this implies $\rho_A \geq (1 - 2\beta^2)b$.

We need a square-integral bound for the short kernel. The decomposition $k = q_a + S_a q_a$, Jensen's inequality for the averaging operator, and its preservation of ν give

$$\int k(d(o, z))^2 \nu(dz) \leq 4 \int q_a(d(o, z))^2 \nu(dz).$$

Integration by parts in the normalized radial element $\frac{1}{2} \sinh r \, dr$ gives

$$\int q_a^2 d\nu = \int_0^a q_a(r) \tanh(r/2) \, dr \leq \frac{1}{2} \int_0^a r \log(a/r) \, dr = \frac{a^2}{8}.$$

Here $q_a(r) = \int_r^a ds / \sinh s \leq \log(a/r)$; the boundary terms vanish, including at zero since $r^2 \log^2 r \rightarrow 0$. Thus

$$\int k(d(o, z))^2 \nu(dz) \leq \frac{a^2}{2}. \quad (51)$$

Fix z and put $P_z(w) = 1 - e^{-\beta k(d(z, w))}$ and $r_z = \frac{\beta^2}{2} (k^2 * b)(z)$. Since $b \leq c$, the preceding bound gives $0 \leq r_z \leq \beta^2 c a^2 / 4 \leq \beta^2$. The inequalities $1 - e^{-x} \geq x - x^2/2$ and $1 - e^{-x} \leq x$ give

$$\int P_z b \, d\nu \geq s(z) - r_z, \quad \int P_z A \, d\nu \leq e^t s(z).$$

For any finite list $p_i \in [0, 1]$, the even Bonferroni inequality is $\prod_i (1 - p_i) \leq 1 - \sum_i p_i + \sum_{i < j} p_i p_j$. For example, it follows by applying the two-term union bound to independent events of probabilities p_i . Apply this inequality to the product in (??); then use the lower bound $\rho_A \geq e^{-d} b$ and the upper bound on the second correlation. The result is

$$\frac{\rho_A(z)}{b(z)} \leq e^{s(z)} \left[1 - e^{-d}(s(z) - r_z) + \frac{1}{2} e^{2t} s(z)^2 \right].$$

All terms are integrable by the estimates already obtained. Since $e^s(1 - s) \leq 1$ for $s \geq 0$ and $1 - e^{-d} \leq d$, this yields

$$\frac{\rho_A(z)}{b(z)} \leq 1 + e^t \left(td + r_z + \frac{1}{2} e^{2t} t^2 \right) \leq 1 + \beta^2 e^\beta \left(\beta e^\beta + 1 + \frac{1}{2} e^{2\beta} \right).$$

For $\beta \leq 1/20$, use $e^\beta \leq 20/19$ and $e^{2\beta} \leq 10/9$ to bound the last coefficient by $5500/3249 < 2$. This proves the upper bound and hence the proposition. $\square$

8.1 Existence and uniqueness of an exact centering

The residue can in fact be removed from the grandcanonical linear field term. The new centering is implicit; its construction does not assume that the original gas is locally neutral.

Theorem 8.3 (An exactly self-consistent finite-range reference). *For every $\lambda > 0$ there is a unique radial function m_λ in the order interval $0 \leq m_\lambda \leq \lambda u$ such that*

$$m_\lambda = \rho_{A_\lambda}, \quad A_\lambda = \lambda u \exp[-\beta(C_a * m_\lambda)]. \quad (52)$$

Both functions are positive and continuous in the open disk and have finite ν -mass and finite C_a -energy. They satisfy the uniform activity estimate

$$0 < m_\lambda \leq A_\lambda \leq c \exp(te^{2t}). \quad (53)$$

In particular, uniformly in λ ,

$$\sup_z \int A_\lambda(w) (1 - e^{-\beta k(d(z, w))}) \nu(dw) \leq te^{te^{2t}} \leq \frac{113918}{2180800} < \frac{1}{19}. \quad (54)$$

With the additional weight $e^{d(z, w)}$ inside the integral, the row sum is at most $e^{2\sqrt{\beta}}/19 < 2/19 < 1$. The maps $\lambda \mapsto m_\lambda$ and $\lambda \mapsto A_\lambda$ are continuous in $L^1(\nu)$ and in the supremum norm.

Proof. Put $G_a = S_a^2 g$, so that $C_a = G_a + h$. The kernel G_a is nonnegative, continuous, and positive semidefinite, and its diagonal is the constant $g(a)$. Positivity of the two by two Gram matrices therefore gives $0 \leq G_a(z, w) \leq g(a)$. Continuity also follows directly by first averaging to obtain the bounded continuous radial function $g(\max\{r, a\})$, and averaging once more. If m is radial and integrable, angular averaging gives $h * m = 0$, with absolute integration justified by the boundedness of $h(z, \cdot)$ for each fixed z . Thus $C_a * m = G_a * m$.

Consider the convex set

$$\mathcal{K}_\lambda = \{m \in L^1(\nu) : m \text{ is radial, } 0 \leq m \leq \lambda u\}.$$

It is weakly compact. To see this without a finite-volume assumption, write $m = \lambda u f$ with $0 \leq f \leq 1$ in the finite measure $u\nu$; its unit order interval is weak-star compact in $L^\infty(u\nu)$, and integration against any bounded function gives its continuous image in the weak topology of $L^1(\nu)$. For $m \in \mathcal{K}_\lambda$ let $A_m = \lambda u e^{-\beta G_a * m}$ and $T(m) = \rho_{A_m}$. The insertion upper bound gives $0 \leq T(m) \leq A_m \leq \lambda u$, and rotation covariance makes $T(m)$ radial. Hence T maps $\mathcal{K}_\lambda$ into itself.

We verify continuity in the indicated weak topology. Weak convergence $m_j \rightarrow m$ implies $(G_a * m_j)(z) \rightarrow (G_a * m)(z)$ at every z , since $G_a(z, \cdot)$ is bounded. Dominated convergence then gives $A_{m_j} \rightarrow A_m$ in $L^1(\nu)$. Finite repulsive gases depend continuously in total variation on their finite activity measures. Indeed, telescope the n activity factors in the partition series: when their masses are bounded by L , the sum of the absolute differences of the unnormalized sector densities is at most $e^L \|A - B\|_1$. The partition functions are at least one, so normalization preserves convergence. Insertion formula (??), whose expectation is of a function in $[0, 1]$, now gives

$$\|\rho_A - \rho_B\|_1 \leq \|A - B\|_1 + \|B\|_1 \text{TV}(Q_A, Q_B).$$

Thus $T(m_j) \rightarrow T(m)$ in L^1 and, in particular, weakly. The Schauder–Tychonoff fixed-point theorem on a compact convex set now gives a fixed point. The preceding argument applies equally to nets, or one can use that the finite measure $u\nu$ has separable L^1 and its bounded weak-star order interval is metrizable.

Uniqueness follows from an entropy identity. If m, n are two fixed points, the logarithm of the density ratio of their reference gases is a constant minus $\beta\eta(G_a * (m - n))$. This linear statistic is integrable, because its summand is bounded and both expected counts are finite. Adding the two relative entropies therefore gives

$$\begin{aligned} D(Q_{A_m} \| Q_{A_n}) + D(Q_{A_n} \| Q_{A_m}) &= \int (m - n) \log(A_m/A_n) d\nu \\ &= -\beta G_a((m - n)\nu, (m - n)\nu) \leq 0. \end{aligned}$$

The energy is nonnegative by positive semidefiniteness and boundedness of G_a , by finite-measure approximation. Relative entropies are nonnegative, so the reference laws coincide and their intensities m, n coincide. This proves uniqueness.

The fixed point is positive by (??): the empty configuration has positive reference probability. Its activity is continuous by the continuity and boundedness of G_a and dominated convergence. For each finite configuration, $z \mapsto \exp[-\beta \sum_{w \in \eta} k(d(z, w))]$ is continuous, with value zero at a coinciding point. Here k is continuous away from zero and tends to infinity at zero. Dominated convergence in (??) proves continuity of m . Finite mass follows from $m, A \leq \lambda u$. The boundary asymptotic of u , with $p > 1$, gives $\int [-\log(1 - |z|^2)] m d\nu < \infty$; the energy justification in Theorem ?? therefore applies.

We next prove parameter continuity, needed to select the bounded branch in the uniform estimate. If $\lambda_j \rightarrow \lambda > 0$, the profiles lie in one weakly compact order interval $0 \leq m \leq \Lambda u$. Every weak limit is a fixed point at λ , by the continuity just proved and the change in the scalar

activity. Uniqueness identifies it with m_λ , and the same argument gives strong L^1 convergence of the intensities and activities. Furthermore

$$\|G_a * (m_j - m)\|_\infty \leq g(a)\|m_j - m\|_1.$$

Consequently the activities converge uniformly, and the pointwise insertion formula, the uniform bound $A_j \leq \Lambda$, and total variation convergence of their laws give uniform convergence of the intensities. This proves both asserted continuities. As $\lambda \downarrow 0$, $\|m_\lambda\|_\infty \leq \lambda$.

It remains to prove the uniform bound. Write $m = m_\lambda$, and set

$$w(z) = \log(\lambda u(z)) - \beta(C * m)(z).$$

For a radial nonnegative integrable charge, angular averaging yields

$$(C * m)(r) = \int -\log \max\{r, |y|\} m(y) \nu(dy) \geq 0.$$

It is finite and continuous and has boundary limit zero. Writing $B = C * m$, differentiation of that one-dimensional formula gives

$$B'(r) = -\frac{2}{r} \int_0^r \frac{s m(s)}{(1-s^2)^2} ds.$$

Thus B is twice continuously differentiable for $r > 0$ and satisfies $\Delta_{\mathbb{H}} B = -m/2$. At zero both $B''(r)$ and $B'(r)/r$ tend to $-m(0)$; the radial Cartesian Hessian therefore extends continuously there, proving the same classical equation at the center. Since $\Delta_{\mathbb{H}} \log u = -p = -\beta c/2$,

$$\Delta_{\mathbb{H}} w = \frac{\beta}{2}(m - c). \quad (55)$$

The insertion Jensen bound and the fixed-point equation imply

$$m \geq A e^{-\beta k * m} = e^w.$$

Also $w \rightarrow -\infty$ at the boundary, so w attains an interior maximum. At that maximum (??) gives $m \leq c$. Hence $e^w \leq c$ everywhere. It follows that

$$m \leq A = e^w e^{\beta k * m} \leq c \exp(\beta M \|m\|_\infty).$$

Put $x(\lambda) = \|m_\lambda\|_\infty / c$. It is continuous, tends to zero with λ , and satisfies $x \leq e^{tx}$. It cannot ever reach 2, because $e^{2t} < 2$ in the present range. Thus $x < 2$ for all activities and then $x \leq e^{2t}$. Substitution in the preceding activity estimate proves (??).

Finally $1 - e^{-\beta k} \leq \beta k$ proves the first inequality in (??). For $0 \leq t \leq 79/1600$,

$$te^{te^{2t}} \leq \frac{t}{1-t/(1-2t)} = \frac{t(1-2t)}{1-3t} \leq \frac{113918}{2180800} < \frac{1}{19}.$$

The first step uses $e^x \leq (1-x)^{-1}$ twice. The last rational function is increasing on this interval, as direct differentiation shows, and its value at $79/1600$ is the displayed fraction. The weighted bound follows from the support condition $d(z, w) < 2a$; here $2\sqrt{\beta} < 1/2$ and $e^{1/2} < 2$. This completes the proof. $\square$

Corollary 8.4 (The exactly centered representation, including counts). *Set $\zeta = \lambda e^{-\beta g(a)/2}$, and use the reference of Theorem ???. Then*

$$\frac{dP_{\zeta, \beta}}{dQ_{A_\lambda}}(\eta) = \frac{\exp[-\frac{\beta}{2} C_a(\eta - m_\lambda \nu, \eta - m_\lambda \nu)]}{E_{Q_{A_\lambda}} \exp[-\frac{\beta}{2} C_a(\eta - m_\lambda \nu, \eta - m_\lambda \nu)]}. \quad (56)$$

In the corresponding imaginary Gaussian representation, the derivative at zero of the centered logarithmic reference functional vanishes: for every bounded real test function f ,

$$\left. \frac{d}{ds} \right|_{s=0} \left[-is\sqrt{\beta} \int m_\lambda f d\nu + \log \Xi_{\text{short}}(A_\lambda e^{is\sqrt{\beta}f}) \right] = 0. \quad (57)$$

The periodic restoring estimate of Proposition ?? applies to this reference uniformly in λ .

Proof. The exact fixed-point activity identity is $\log(\lambda u) = \log A_\lambda + \beta C_a * m_\lambda$. Use it in (??) and complete the square, as in Theorem ?. This proves the identity in every count sector, including count zero. The finite energies and strictly positive denominator are justified exactly as there, using $m_\lambda \leq \lambda u$. The logarithmic derivative of a grand partition function is its intensity paired with the activity perturbation. Since $\rho_{A_\lambda} = m_\lambda$, the two terms in (??) cancel exactly. All differentiations are justified by Poisson domination of the finite partition series for bounded f . The row bound (??) proves the final assertion. $\square$

Corollary 8.5 (Matching the reference mass to each canonical count). *The mass $\mathcal{M}(\lambda) = \int m_\lambda d\nu$ is continuous and strictly increasing from zero to infinity. Thus for every $N \geq 1$ there is a unique $\hat{\lambda}_N$ for which $\mathcal{M}(\hat{\lambda}_N) = N$. Conditioning (??) at this parameter on count N gives an exact representation of the original canonical gas.*

Proof. Continuity follows from the L^1 continuity in the theorem, and $\mathcal{M}(\lambda) \leq \lambda \int u d\nu$ gives its zero limit. For two parameters $\lambda > \kappa$, the same entropy calculation as in the uniqueness proof gives

$$\begin{aligned} D(Q_{A_\lambda} \| Q_{A_\kappa}) + D(Q_{A_\kappa} \| Q_{A_\lambda}) + \beta C_a((m_\lambda - m_\kappa)\nu, (m_\lambda - m_\kappa)\nu) \\ = (\log \lambda - \log \kappa)(\mathcal{M}(\lambda) - \mathcal{M}(\kappa)). \end{aligned}$$

Every term on the left is nonnegative. Equality of the two masses would imply equality of the reference laws and hence of their intensities. Their activities would also coincide, since the ratio of the singleton-sector density to the vacuum probability equals the activity density. Their activity equations would then force $\lambda = \kappa$. Thus the masses are strictly increasing. At the origin,

$$A_\lambda(0) = \lambda e^{-\beta(G_a * m_\lambda)(0)} \geq \lambda e^{-\beta g(a)\mathcal{M}(\lambda)}.$$

The uniform activity upper bound implies $\mathcal{M}(\lambda) \geq [\log \lambda - \log(cc^{te^{2t}})]/[\beta g(a)]$, so the mass diverges. The intermediate value theorem gives the unique matching parameter. Every count sector has positive probability, so conditioning the exact identity is legitimate. $\square$

Remark 8.6 (The gain and the remaining screening problem). Exact centering removes one previously explicit obstruction: the grandcanonical imaginary field action now has no linear term, while its finite-range reference still has a uniform row norm below $1/19$. The reference retains all particle-number weights. This does not remove the other periodic wells, prove a lower bound for oscillatory normalizing integrals, or transfer reference mixing through the long-range quadratic tilt. It also does not assert $m_\lambda \rightarrow c$, nor that the actual gas has intensity m_λ . After conditioning the reference on count N , its intensity need not equal $m_{\hat{\lambda}_N}$; matching the expected total mass alone does not preserve the grandcanonical linear cancellation for every spatial test. The canonical constant phase remains unchanged in modulus. These distinctions prevent the exact centering theorem from being read as a proof of the intensity conjecture.

9 Exact self-consistent centering at fixed particle number

The grandcanonical construction has a fixed-count counterpart. It removes the linear field term at each finite N , but does not supply the uniform reference estimates proved above for the grandcanonical gas. Throughout this section $0 < \beta < 2$, $a > 0$, $N \geq 1$, and $u(z) = (1 - |z|^2)^p$, where $p = 2 - \beta/2 > 1$.

Theorem 9.1 (Canonical centering). *There is a unique finite-energy self-consistent canonical reference whose activity, up to a multiplicative constant, is*

$$A_N(z) = u(z) \exp[-\beta(C_a * b_N)(z)]$$

and whose intensity relative to ν is b_N . Here the reference Q_N^ has exactly N particles and pair interaction βk_a . The profile is radial and satisfies*

$$\int b_N d\nu = N, \quad e^{-\beta g(a)N} u \leq A_N \leq u.$$

More precisely, finite energy in the uniqueness assertion means that $b_N \nu$ has an absolutely defined Hilbert-space mean for the feature map of C_a . The original canonical disk gas obeys the exact identity

$$\frac{dP_{N,\beta}}{dQ_N^*}(\eta) = \frac{\exp[-\frac{\beta}{2}C_a(\eta - b_N \nu, \eta - b_N \nu)]}{E_{Q_N^*} \exp[-\frac{\beta}{2}C_a(\eta - b_N \nu, \eta - b_N \nu)]}. \quad (58)$$

Proof. Since C_a is a continuous positive semidefinite kernel on the disk, it has a real separable Hilbert feature space H : take the span of symbols Φ_z , give them inner products $\langle \Phi_z, \Phi_w \rangle = C_a(z, w)$, quotient by the null space, and complete. Continuity of the kernel makes $z \mapsto \Phi_z$ continuous and a countable dense set of disk points spans a dense subspace. In particular,

$$\|\Phi_z\|^2 = D(z) := g(a) - \log(1 - |z|^2).$$

Let Q_N^0 be the canonical short-range gas with activity $u\nu$, and set $X(\eta) = \sum_{z \in \eta} \Phi_z \in H$. Its partition function is positive and finite because $k_a \geq 0$ and $\int u d\nu = (p-1)^{-1}$. For every $t > 0$,

$$E_{Q_N^0} e^{t\|X\|} < \infty. \quad (59)$$

Indeed $\|X\| \leq \sum_z \sqrt{D(z)}$, so the unnormalized expectation is bounded by a constant times $[\int u(z) e^{t\sqrt{D(z)}} \nu(dz)]^N$. Writing $s = -\log(1 - |z|^2)$ evaluates the latter integral as

$$\int_0^\infty e^{-(p-1)s+t\sqrt{g(a)+s}} ds < \infty.$$

Consider the functional on H

$$J_N(h) = \frac{1}{2}\|h\|^2 + \frac{1}{\beta} \log E_{Q_N^0} e^{-\beta\langle h, X \rangle}.$$

The exponential moments justify its finiteness, norm continuity, and Fréchet differentiation on every bounded set of H . For example, the difference quotients and their first two derivatives are dominated by a polynomial in $\|X\|$ times $e^{\beta R\|X\|}$ on $\|h\| \leq R$; these functions are integrable by (??). If $Q_{N,h}$ is the probability measure obtained by the displayed exponential tilt, then

$$\begin{aligned} \nabla J_N(h) &= h - E_{Q_{N,h}} X, \\ D^2 J_N(h)[v, v] &= \|v\|^2 + \beta \text{Var}_{Q_{N,h}} \langle v, X \rangle \geq \|v\|^2. \end{aligned}$$

Thus J_N is strongly convex. Jensen's inequality gives

$$J_N(h) \geq \frac{1}{2}\|h\|^2 - \langle h, E_{Q_N^0} X \rangle,$$

so it is coercive. A minimizing sequence is bounded and therefore has a weakly convergent subsequence in the Hilbert space. A norm-continuous convex functional is weakly lower semicontinuous, so the weak limit is a minimizer. Strong convexity makes this minimizer h_N unique, and its gradient equation reads

$$h_N = E_{Q_{N,h_N}} X. \quad (60)$$

Put $Q_N^* = Q_{N,h_N}$ and let b_N be its intensity density. All first moments of $\sum \sqrt{D}$ remain finite under the tilt, by the same exponential-moment argument. Consequently

$$h_N = \int \Phi_z b_N(z) \nu(dz), \quad \langle \Phi_z, h_N \rangle = (C_a * b_N)(z),$$

where the first integral is a Bochner integral and the second is absolutely convergent by Cauchy–Schwarz. This proves self-consistency. Conversely every self-consistent reference in the stated finite-energy class defines a vector h with $h = E_{Q_{N,h}} X$ and hence $\nabla J_N(h) = 0$; uniqueness of the minimizer gives uniqueness of both the reference and its intensity.

Rotations act unitarily on H by $\Phi_z \mapsto \Phi_{e^{i\theta}z}$. The kernel and Q_N^0 are rotation invariant, so J_N is invariant under these unitaries. Its unique minimizer is invariant, proving radially of the reference and its intensity. For a radial background the harmonic part of C_a has zero convolution, and $C_a * b_N = G_a * b_N$, where $G_a = S_a^2 g$. The bounded positive semidefinite kernel G_a satisfies $0 \leq G_a(z, w) \leq g(a)$: its diagonal is $g(a)$ and the upper bound follows from Cauchy–Schwarz in its feature space. These facts justify the displayed activity bounds because $\int b_N = N$. The harmonic cancellation is legitimate even at the disk boundary: for fixed z its kernel is bounded as a function of w , and the background has finite mass.

Finally, with $h = h_N$ and $X = X(\eta)$, completing the square gives

$$e^{-\frac{\beta}{2}\|X\|^2} \prod_{z \in \eta} u(z) = e^{\frac{\beta}{2}\|h\|^2} e^{-\frac{\beta}{2}\|X-h\|^2} \prod_{z \in \eta} u(z) e^{-\beta \langle \Phi_z, h \rangle}.$$

Multiplying by the short-range interaction and using the exact smearing identity for the disk gas proves (??). Its denominator is strictly positive and at most one, because the energy is a finite squared norm. $\square$

Corollary 9.2 (Cancellation of every canonical linear phase). *Let $\mathcal{Z}_N(A)$ be the short-range canonical partition functional. For every bounded real ψ , the function*

$$F_N(t; \psi) = e^{-it \int b_N \psi d\nu} \frac{\mathcal{Z}_N(A_N e^{it\psi})}{\mathcal{Z}_N(A_N)}$$

satisfies $F_N(0; \psi) = 1$ and $F'_N(0; \psi) = 0$. If W is an isonormal Gaussian process on H , the exact normalizer in (??) is

$$E_W \left[e^{-i\sqrt{\beta}W(h_N)} \frac{\mathcal{Z}_N(A_N e^{i\sqrt{\beta}W(\Phi_\cdot)})}{\mathcal{Z}_N(A_N)} \right]. \quad (61)$$

For a constant ψ , one has $F_N(t; \psi) = 1$ for every real t .

Proof. The partition ratio is $E_{Q_N^*} e^{it\eta(\psi)}$; centering by its mean gives the derivative assertions. A constant statistic equals $N\psi$ deterministically, proving the last claim. The Gaussian characteristic function gives

$$E_W e^{i\sqrt{\beta}W(X-h_N)} = e^{-\frac{\beta}{2}\|X-h_N\|^2}.$$

Average this identity under Q_N^* to obtain (??). Fubini is justified because the integrand before either expectation has modulus one. A jointly measurable field $W(\Phi_z)$ can be used: expand in an orthonormal basis of the separable feature space, take Gaussian partial sums, and choose their measurable limit in local product L^2 ; its covariance is C_a . Values on the resulting area-null exceptional set do not affect any of the absolutely continuous partition integrals. $\square$

The theorem does not provide bounds uniform in N on short-range influences or on the positive normalizer (??). The grandcanonical insertion identity used in the activity bootstrap compares the reference with itself after insertion. At fixed count, insertion instead involves an $(N - 1)$ -particle reference, whose intensity need not be b_N ; the same bootstrap therefore does not follow. Furthermore the constant phase is exactly flat, so a canonical analogue of the grandcanonical periodic coercivity estimate with a strictly positive penalty for all constant phases is false. Exact centering removes the linear residue; it does not remove these finite-volume normalization and comparison problems.

10 What has been removed from the boundary problem

Theorems ??–?? are actual fixed-temperature estimates. They control all count and position events for the specified perturbations, with constants uniform in N . The force reconstruction also handles all distant particles together. In particular, the Green force cannot retain an extra random component at infinity for $0 < \beta \leq 2$ after its configuration-measurable finite truncations have been retained. The independent radial theorem controls a further actual exterior channel at every temperature.

At the determinantal benchmark, Theorem ?? goes further: the entire far exterior loses influence on the whole local law at rate $O_r(1 - R^2)$, uniformly in N . Reversing the conditioning reduces the estimate to the distant excess count created by a fixed inner hole. Proposition ?? formulates the same reduction for other temperatures under explicit stochastic-domination and hole-response hypotheses. The explicit discrete example in Proposition ?? shows why Lorentzian structure alone does not supply that domination. It is not a counterexample for the disk interaction; the latter still has additional structure which might yield a comparison.

These results leave two kinds of transmitted data unresolved. The first is the harmonic contribution in the decomposition $-\log|a - z| = g(a, z) - \log|1 - a\bar{z}|$. The second is the scalar normalization which governs local particle-number weights. Given an exterior Y and a prescribed count m , the exact conditional density in a region U is

$$\frac{1}{Z_{U,m,Y}} |\Delta_m(\mathbf{z})|^\beta \exp \left\{ \beta \sum_i \sum_{y \in Y} \log |z_i - y| \right\} \prod_i \mathbf{1}_U(z_i) \mu(dz_i).$$

Adding a constant c_0 to the exterior potential has no effect in one fixed sector, but multiplies its unnormalized weight by $e^{\beta m c_0}$ when sectors are mixed. Force reconstruction therefore does not determine the relative count probabilities.

The exact reference representation in Section ?? addresses part of this normalization problem: it retains every count factor, constructs the reference at every activity and at every canonical mass, and proves a small short-range comparison norm without assuming the actual intensity. It still requires an estimate on the normalized long-range quadratic tilt under local changes. Dropping that tilt because its energy is nonnegative would be invalid. Conditioning the short-range reference on its total count also introduces a dependency that the grandcanonical row-sum estimate does not automatically control. For $\beta \leq 1/20$, the complex cluster expansion additionally proves an explicit periodic restoring bound for the exact imaginary-field integrand. The bound is quadratic in each small-field well, but does not control other wells or cancellation in the normalized oscillatory integral.

The new self-consistent construction in Theorem ?? removes the imaginary linear residue for $\beta \leq 1/20$ while preserving both the Mayer bound and a uniformly subcritical distance-weighted influence norm. Theorem ?? separately removes every linear phase at each fixed N for $0 < \beta < 2$. Its proof uses a strongly convex Hilbert-space functional. The canonical theorem does not give a reference influence estimate uniform in N , and the constant canonical phase remains

exactly flat. Neither construction provides the normalized long-range comparison needed for the original local particle law.

The profile b_λ constructed there converges locally to $4/\beta - 1$. It is a deterministic balancing profile. It is not asserted to be the intensity of either the interacting short-range reference or the actual gas. Consequently the construction does not prove

$$\mathbb{E}\Xi_\beta(dz) = (4/\beta - 1)\nu(dz).$$

The Green-force estimate likewise concerns a potential modulo constants and cannot identify this coefficient by itself. These are remaining estimates for the original model, not counterexamples to its conjectured limit.

The susceptibility argument gives no comparable theorem for $\beta > 2$: the negative-type property used at its base is specific to exponents at most two. Nor does the two-to-one coupling in the appendix imply a coupling between every pair of successive particle numbers. The simultaneous local linear response at 2 gives an additional perturbative input, while higher connected temperature responses and a uniform remainder remain uncontrolled.

A An exact non-determinantal transfer from two particles to one

The next result restores a genuine count-transfer coupling below two. Its scope is two interior particles versus one; the extension to arbitrary interior counts is not established here.

Proposition A.1 (Deletion coupling for $0 < \beta \leq 2$). *Let σ be a finite nonatomic measure of compact support, let $y \in \mathbb{R}^2$, and let $0 < \beta \leq 2$. Compare the unordered pair law*

$$P_2(dz dw) = \frac{|z - w|^\beta \sigma(dz) \sigma(dw)}{\iint |u - v|^\beta \sigma(du) \sigma(dv)}$$

with the singleton law

$$Q_y(dz) = \frac{|z - y|^\beta \sigma(dz)}{\int |u - y|^\beta \sigma(du)}.$$

There is a coupling $(\{Z, W\}, X)$ with these marginals such that $X \in \{Z, W\}$ almost surely. In particular, for any finite exterior Y and observation region U , the actual disk-gas conditional law with two particles in U and exterior Y dominates the conditional law with one particle in U and exterior $Y \cup \{y\}$.

Proof. Normalize σ to have mass one and use the Hilbert embedding of Lemma ???. Put $h = \Phi(y)$, let m and $V > 0$ be the mean and variance of Φ under σ . For a Borel set B with $b = \sigma(B) \in (0, 1)$, put $a = 1 - b$. Let u, v be the conditional Hilbert means on B, B^c , and S, T their conditional variances. Thus

$$m = bu + av, \quad V = bS + aT + ab\|v - u\|^2.$$

The probabilities of interest are

$$Q_y(B) = \frac{b(S + \|u - h\|^2)}{V + \|m - h\|^2}, \quad P_2(\{Z, W\} \subset B) = \frac{b^2 S}{V}.$$

We claim the first is at least the second. Put $x = u - h$ and $d = v - u$. The numerator after multiplying by the positive denominators and cancelling b is

$$\begin{aligned} & V(S + \|x\|^2) - bS(V + \|x + ad\|^2) \\ &= a \left[b(\|d\|^2 \|x\|^2 - 2S\langle x, d \rangle + S^2) + T\|x\|^2 + aST \right] \geq 0. \end{aligned}$$

The last inequality follows from Cauchy–Schwarz; the expression in parentheses is at least $(\|d\|\|x\| - S)^2$. Sets of σ -mass zero or one satisfy the same inequality directly. Replacing B by its complement gives, for every Borel A ,

$$Q_y(A) \leq P_2(\{Z, W\} \cap A \neq \emptyset). \quad (62)$$

For completeness, these Hall inequalities give the required continuum coupling as follows. Partition a compact set containing the support into finitely many Borel cells of diameter at most ε . The right states are unordered pairs of cells, including equal cells; a left cell can be matched to a pair precisely when it belongs to that pair. Inequalities (??), applied to unions of cells, are the finite capacitated Hall inequalities. The max-flow/min-cut theorem therefore supplies a coupling of the cell-valued singleton and pair, supported on this incidence relation. Lift the finite coupling using the conditional laws within the cells, and within the pair states. It has the exact original marginals and satisfies $\min(|X - Z|, |X - W|) \leq \varepsilon$ almost surely. Compactness gives a weakly convergent subsequence as $\varepsilon \downarrow 0$. The marginals persist, and continuity of $\min(|x - z|, |x - w|)$ forces its limiting value to be zero almost surely. This is the desired deletion coupling.

For the disk-gas assertion, take $\sigma(dz) = \mathbf{1}_U(z) \prod_{a \in Y} |z - a|^\beta dA(z)$. The two displayed densities are exactly the two conditional laws. $\square$

Remark A.2. The deleted point has intensity measure $\rho_{P_2} - Q_y$, independently of the choice of deletion coupling, because the difference consists of one point. This proves a count-sensitive transport statement for the original interaction throughout $0 < \beta \leq 2$. It supplies no distance decay by itself and does not justify iterating the coupling through sectors with three or more particles.

B A bounded response of the entire local law at temperature two

Here is an actual disk-gas estimate which includes every local count sector. It concerns the derivative at $\beta = 2$, and does not assume that a general-temperature limiting process exists. Throughout this section, $\mu = dA/\pi$, $W = B_r$ with $0 < r < 1$, and

$$S_N = \sum_{i < j} \log |z_i - z_j|, \quad dP_{N,\beta} = Z_N(\beta)^{-1} e^{\beta S_N} d\mu^N.$$

Let $P_{N,\beta}^W$ denote its law restricted to W . For a finite signed measure σ , write $\|\sigma\|_{\text{var}} = |\sigma|(\Omega)$; the probability total variation distance is half this norm.

Theorem B.1 (Uniform whole-law linear response). *For each $r < 1$, the finite signed measures*

$$\dot{P}_N^W = \partial_\beta P_{N,\beta}^W \Big|_{\beta=2}$$

exist in variation norm, have total mass zero, and converge in variation norm as $N \rightarrow \infty$. In particular, there is $C_r < \infty$ such that

$$\sup_N \|\dot{P}_N^W\|_{\text{var}} \leq C_r, \quad |\text{Cov}_{N,2}(F(\Xi_N|_W), S_N)| \leq C_r \|F\|_\infty$$

for every bounded measurable local observable F . The convergence is simultaneous over $\|F\|_\infty \leq 1$, including all local count events.

This supplies a uniformly bounded local response to an extensive pair-energy perturbation. It does not supply an interval of temperatures: a uniform bound for a derivative at one temperature does not control its nonlinear remainder uniformly in N .

B.1 Kernel estimates with the collision singularities removed

At $\beta = 2$ the correlation kernel, relative to μ , is

$$K_N(x, y) = \sum_{a=1}^N a(x\bar{y})^{a-1}, \quad K(x, y) = (1 - x\bar{y})^{-2}.$$

Write $\ell(x, y) = \log |x - y|$ and $g_N(x) = \frac{1}{2} \sum_{a=1}^N |x|^{2a}/a$, $g(x) = -\frac{1}{2} \log(1 - |x|^2)$.

Lemma B.2. *One has*

$$\iint |\ell| |K_N - K| d\mu^2 \longrightarrow 0, \quad J := \sup_N \iint |\ell| |K_N| d\mu^2 < \infty.$$

Also $\int g d\mu = 1/2$, and

$$L := \sup_{z \in \mathbb{D}} \int_{\mathbb{D}} |\log |z - x|| d\mu(x) \leq \frac{1}{2} + \log 2.$$

Proof. The last two assertions follow by integration: for the negative logarithm translate into the unit disk about z , and for its positive part use $|z - x| \leq 2$. Put $u = x\bar{y}$. The exact finite-series identity gives

$$|K_N - K| \leq \frac{|u|^N}{|1 - u|^2} + \frac{N|u|^N}{|1 - u|}.$$

Where $\min(|x|, |y|) \leq 1/2$, denominators are bounded below and the logarithmic singularity is uniformly integrable, so the integral tends to zero. On the complement, put $v = |x||y|$ and let $\theta \in [-\pi, \pi]$ be the angular difference. Uniformly there,

$$|1 - u|^2 \geq c((1 - v)^2 + \theta^2), \quad |\ell(x, y)| \leq C(1 + |\log |\theta||).$$

Consequently the angular integrals with denominators $|1 - u|^2$ and $|1 - u|$ are bounded respectively by

$$C \frac{1 + |\log(1 - v)|}{1 - v}, \quad C(1 + |\log(1 - v)|^2).$$

The radial integration identity

$$\int_0^1 \int_0^1 f(rs) r dr s ds = \int_0^1 f(v) v (-\log v) dv$$

shows that the first bound without v^N is integrable, so its version with v^N tends to zero by dominated convergence. For the second bound, split at $v = 1/2$; on the upper half use $-\log v \leq 2(1 - v)$ and $v^N \leq e^{-N(1-v)}$. Substituting $t = N(1 - v)$ bounds the integral by $C(1 + \log^2(N+1))/N$, which tends to zero. The same integrable first bound proves $\iint |\ell| |K| < \infty$. The assertions follow. $\square$

Fix $k \leq N$ and distinct roots $\mathbf{z} = (z_1, \dots, z_k) \in W^k$. Set $G_N = [K_N(z_i, z_j)]$, $D_N = \det G_N$, and

$$Q_N(x, y) = K_N(x, \mathbf{z}) G_N^{-1} K_N(\mathbf{z}, y), \quad q_N(x) = Q_N(x, x).$$

These formulas are well defined: evaluations at k distinct points are linearly independent on the polynomials of degree less than N , by polynomial interpolation. Q_N is a rank- k orthogonal projection, and the reduced Palm kernel is $K_N - Q_N$. These statements also follow directly by taking the Schur complement in the correlation determinant.

Lemma B.3 (Gram cofactor bounds). Put $M = (1 - r^2)^{-2}$ and $A = (1 - r)^{-2}$. Uniformly in $N, k, \mathbf{z}$ as above and $x, y \in \mathbb{D}$,

$$\begin{aligned} 0 &\leq D_N \leq M^k, \\ D_N |Q_N(x, y)| &\leq k^2 M^{k-1} A^2, \\ 0 &\leq D_N \{q_N(x)q_N(y) - |Q_N(x, y)|^2\} \leq 4 \binom{k}{2}^2 M^{k-2} A^4. \end{aligned}$$

The last expression is zero when $k = 1$. These bounds remain uniform when the roots approach collisions.

Proof. The diagonal Gram entries are at most M and the entries $K_N(x, z_i)$ have absolute value at most A . For any two equally sized index sets I, J , the Gram-minor Cauchy–Schwarz inequality gives

$$|\det(G_N)_{I,J}|^2 \leq \det(G_N)_{I,I} \det(G_N)_{J,J} \leq M^{2|I|}.$$

This inequality follows by writing the minor as the inner product of the corresponding exterior products of Gram vectors. The first assertion is Hadamard’s inequality. Expanding $D_N Q_N$ using the adjugate of G_N gives k^2 terms, each bounded by $M^{k-1} A^2$.

For the last assertion, form the k by 2 matrix V of evaluations at x, y . Its left side is $D_N \det(V^* G_N^{-1} V)$. Cauchy–Binet and the complementary-minor formula for G_N^{-1} express this as a sum over pairs of two-element index sets. The complementary Gram minor is bounded by M^{k-2} ; each two by two evaluation determinant has absolute value at most $2A^2$. There are $\binom{k}{2}^2$ terms. Positivity follows because $V^* G_N^{-1} V$ is positive semidefinite. $\square$

B.2 The exact derivative and its all-order envelope

Let $R_{N,\beta}^{(k)}$ be the factorial correlation density relative to μ^k . The angular mean identity for the logarithm gives

$$h_N(x) := \int \ell(x, y) K_N(y, y) d\mu(y) = -\frac{1}{2} \sum_{a=1}^N \frac{1}{a} + g_N(x). \quad (63)$$

Indeed, for a single term $a|y|^{2a-2}$, radial integration of $\log \max(|x|, |y|)$ gives $(|x|^{2a} - 1)/(2a)$.

Write $L_{\mathbf{z}}(x) = \sum_{i=1}^k \ell(z_i, x)$ and $\Delta(\mathbf{z}) = \prod_{i < j} (z_i - z_j)$. Define

$$\begin{aligned} B_N(\mathbf{z}) &= \sum_i g_N(z_i) - \int q_N g_N d\mu - \int L_{\mathbf{z}} q_N d\mu \\ &\quad + \frac{1}{2} \iint \ell(x, y) \{q_N(x)q_N(y) - |Q_N(x, y)|^2\} d\mu^2 \\ &\quad + \iint \ell(x, y) \operatorname{Re}\{K_N(x, y) \overline{Q_N(x, y)}\} d\mu^2. \end{aligned} \quad (64)$$

Then the exact identity is

$$\dot{R}_N^{(k)}(\mathbf{z}) := \partial_\beta R_{N,\beta}^{(k)}(\mathbf{z}) \Big|_2 = D_N \{\log |\Delta(\mathbf{z})| + B_N(\mathbf{z})\}. \quad (65)$$

Here is a derivation to make the normalization cancellation explicit. Differentiating the root-density integral gives

$$\frac{\dot{R}_N^{(k)}}{D_N} = \log |\Delta| + E_{N,2}^{\mathbf{z}} \left[S_{N-k} + \sum_{y \in Y} L_{\mathbf{z}}(y) \right] - E_{N,2} S_N.$$

Expand the one- and two-point densities of the Palm kernel $K_N - Q_N$. The difference on the right, apart from $\log |\Delta|$, is

$$\begin{aligned} \sum_i h_N(z_i) - \int q_N h_N d\mu - \int L_{\mathbf{z}} q_N d\mu + \frac{1}{2} \iint \ell(q_N q_N - |Q_N|^2) d\mu^2 \\ + \iint \ell \operatorname{Re}(K_N \overline{Q_N}) d\mu^2. \end{aligned}$$

Since $\int q_N d\mu = k$, the divergent harmonic-number constant in (??) cancels exactly. This proves (??)–(??). All finite- N differentiations are justified by integrability of logarithms times a strictly positive Vandermonde power in a neighborhood of $\beta = 2$.

Proposition B.4. *For each $r < 1$ there are finite constants c_r, C_r such that, for every $k \geq 1$ and $N \geq k$,*

$$a_{N,k} := \int_{W^k} |\dot{R}_N^{(k)}| d\mu^k \leq c_r (k+1)^4 C_r^k.$$

For each fixed k , the functions $\dot{R}_N^{(k)}$ converge in $L^1(W^k, \mu^k)$ as $N \rightarrow \infty$.

Proof. Lemmas ?? and ??, applied term by term in (??), give

$$D_N |B_N| \leq k M^k g(r) + k^2 M^{k-1} A^2 \left(\frac{1}{2} + kL + J\right) + 2 \binom{k}{2}^2 M^{k-2} A^4 L.$$

Call this upper bound $F_{r,k}$. If $v = \mu(W) = r^2$, then

$$a_{N,k} \leq v^k F_{r,k} + M^k \binom{k}{2} v^{k-2} \iint_{W^2} |\ell(x, y)| d\mu^2.$$

For $k = 1$ the second term is zero, and for $k \geq 2$ its double integral is at most vL . This proves the claimed exponential times polynomial bound after enlarging constants depending only on r .

For a fixed distinct root tuple, G_N converges to its positive-definite infinite Bergman Gram matrix. Positivity follows again by testing the evaluation functionals against interpolation polynomials. Also $K_N(x, z_i) \rightarrow K(x, z_i)$ uniformly for $|x| \leq 1$, so $Q_N \rightarrow Q$ uniformly on $(\mathbb{D})^2$ for this tuple. Thus every term in (??) converges: use $g_N \leq g$, integrability of the logarithms, and Lemma ?? for the final two-variable term. The bound just obtained, together with $M^k \sum_{i < j} |\ell(z_i, z_j)|$, is an integrable pointwise majorant on W^k . Dominated convergence proves L^1 convergence. The collision set has μ^k measure zero. $\square$

B.3 Passing from every correlation to the entire local law

Proof of Theorem ??. For finite N , differentiability of $P_{N,\beta}$ in variation norm follows by differentiating its density in L^1 . Its derivative is $(S_N - E_{N,2} S_N) P_{N,2}$, and restriction to W is a contraction for variation norm.

For a second, more useful expression, let $j_{N,\beta,m}$ be the Janossy density in W : the probability of exactly m unordered points in W is represented by $j_{N,\beta,m} d\mu^m / m!$. The empty sector uses $\mu^0 = 1$. Finite inclusion–exclusion gives

$$j_{N,\beta,m}(\mathbf{x}) = \sum_{\ell=0}^{N-m} \frac{(-1)^\ell}{\ell!} \int_{W^\ell} R_{N,\beta}^{(m+\ell)}(\mathbf{x}, \mathbf{y}) d\mu^\ell(\mathbf{y}).$$

Set $\dot{R}_N^{(k)} = 0$ for $k > N$, and $\dot{R}_N^{(0)} = 0$. Differentiate this finite sum and use the triangle inequality. Regrouping by $k = m + \ell$ yields

$$\|\dot{P}_N^W\|_{\text{var}} = \sum_{m \geq 0} \frac{1}{m!} \int_{W^m} |\dot{j}_{N,m}| d\mu^m \leq \sum_{k \geq 1} \frac{2^k}{k!} a_{N,k} < C_r. \quad (66)$$

The last inequality is uniform by Proposition ??.

Apply the same inequality to the difference of the derivatives for N and N' . For every fixed k , the L^1 difference of the correlation derivatives tends to zero. Its summand is dominated by twice the summable envelope in (??). Hence $\|\dot{P}_N^W - \dot{P}_{N'}^W\|_{\text{var}} \rightarrow 0$. Completeness of the space of finite signed measures in variation norm gives the stated limit. Its total mass is zero by continuity of total mass. Finally, integrating F against the derivative gives the covariance formula and its uniform bound. $\square$

Remark B.5 (What is still missing). The uniform all-order bound above refers to correlation order k , not temperature derivative order. It closes the passage from fixed correlations to all local events at first order. A fixed interval near 2 still requires a uniform nonlinear remainder, or summable bounds on all temperature derivatives. Neither follows from this theorem. For example, an additional term proportional to $\tanh(N(\beta - 2)^2)$ has zero first derivative at 2 for every N , while remaining nonzero at every fixed other temperature as $N \rightarrow \infty$.

References

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